READING 13

ANALYSIS OF DIVIDENDS AND SHARE REPURCHASES

EXAM FOCUS

The focus of the Level II exam is valuation, so pay close attention to the theories that explain how dividend policy affects company value and the signals investors get from dividend changes. Payout policy is broader than dividend policy as it includes other means (e.g., special dividends, stock repurchases, etc.) by which companies can pay out cash to stockholders. In recent years, firms have announced plans to repurchase record numbers of shares, making this an important and timely topic. You should have a basic understanding of the factors that affect a firm's payout policy and be able to analyze the sustainability of dividends using coverage ratios.

MODULE 13.1: THEORIES OF DIVIDEND POLICY

Video covering this content is available online.

LOS 13.a: Describe the expected effect of regular cash dividends, extra dividends, liquidating dividends, stock dividends, stock splits, and reverse stock splits on shareholders' wealth and a company's financial ratios.

Types of dividends include the following:

1. **Regular cash dividends:** Periodic dividend payments made in cash are known as regular cash dividends. Generally, companies strive for stability in their regular dividends—increasing them slowly and refraining from any reductions. Stable or increasing dividends paid regularly are perceived as a sign of consistent (and growing) profitability.

   Frequency of dividend payments vary globally: U.S. and Canadian companies typically pay dividends quarterly, while European and Asian companies typically pay dividends semiannually and annually, respectively.

2. **Extra or special (irregular) dividends:** A cash dividend supplementing regular dividends, or a dividend of a company that normally does not pay dividends, is known as a special dividend. Special dividends are paid under unusual circumstances (e.g., when the company spins off a division) under the expectation that the dividend is not recurring. Extra dividends may be paid if the company had an especially profitable year but does not want to commit to a higher *ongoing* regular dividend payment.
3. **Liquidating dividend:** This is paid by a company when the whole firm or part of the firm is sold, or when dividends in excess of cumulative retained earnings are paid (resulting

in a reduction of stated capital). A liquidating dividend is considered to be a return *of* capital as opposed to a return *on* capital.

4. **Stock dividend:** A non-cash dividend paid in the form of additional shares is known as a stock dividend. After payment of a stock dividend, shareholders have more shares and the cost per share will be lower (while the total cost basis remains the same). Shareholders' proportionate ownership of the company does not change, because every shareholder receives the same percentage stock dividend. Shareholders are usually not taxed on stock dividends. Because the market value of the company is unchanged, the market price per share declines, leaving the shareholders with no net gain.

   Imagine a company earning $100,000 annually that has 100,000 shares outstanding with a market price per share of $10. Current EPS and P/E are then $1.00 and 10, respectively. Suppose that this company then declares a 10% stock dividend; the company now has 110,000 shares outstanding and earnings per share (EPS) of $0.90909. Applying the original P/E multiple of 10, the stock price should now be $9.091. Hence, a holder of 110 shares should have holdings valued at $1000—the same as before the stock dividend.

   Companies consider stock dividends desirable because they encourage long-term investing and, hence, may reduce the cost of equity capital. Stock dividends also help increase a stock's float, and therefore, its liquidity. Stock dividends may also be used to decrease the market price of a stock to a desirable trading range that attracts more investors (high stock prices coupled with a minimum order size tend to limit ownership by small retail investors). Companies that continue to pay the same regular cash dividend per share following a stock dividend have effectively increased their cash dividend. However, companies that pay out the same total *amount* of cash dividends as before (i.e., the same payout ratio) effectively *decrease* the dividend per share (though the dividend yield would be unchanged, because dividends and price decrease by same percentage following the stock dividend).

   The popularity of stock dividends varies by market. For example, stock dividends are very popular in China.

5. **Stock splits:** These are similar to stock dividends (non-cash) but generally larger in size. A two-for-one stock split is the same as a 100% stock dividend. Reverse stock splits (much less common) reduce the number of shares outstanding and increase the price per share. The intent of a reverse stock split is similar to that of a regular stock split—to bring the market price of the stock within a desirable range, but in this case the firm is trying to attract institutional investors and mutual funds that shun low-priced stocks. Reverse stock splits are less common in Asia (e.g., reverse stock splits were illegal in Japan until 2001).

Accounting Issues

Cash dividend payments reduce cash as well as stockholders' equity. This results in a lower quick ratio and current ratio, and higher leverage (e.g., debt-to-equity and debt-to-asset) ratios. Conversely, stock dividends (and stock splits or reverse stock splits) leave a company's capital structure unchanged and do not affect any of these ratios. In the case of a stock dividend, a decrease in retained earnings (corresponding to the value of the stock dividend) is offset by an increase in contributed capital, leaving the value of total equity unchanged. A stock split or reverse stock split does not affect the book value of the equity, nor does it affect the tax cost basis for the shareholders.

LOS 13.b: Compare theories of dividend policy and explain implications of each for share value given a description of a corporate dividend action.

Dividend irrelevance. Merton Miller and Franco Modigliani (MM) maintain that *dividend policy* is irrelevant, as it has no effect on the price of a firm's stock or its cost of capital. MM's argument of dividend irrelevance is based on their concept of *homemade dividends*. Assume, for example, that you are a stockholder and you don't like the firm's dividend policy. If the firm's cash dividend is too big, you can just take the excess cash received and use it to buy more of the firm's stock. If the cash dividend you received was too small, you can sell a little bit of your stock in the firm to get the cash flow you want. In either case, the combination of the value of your investment in the firm and your cash in hand will be the same.

You should note that the dividend irrelevance theory holds only in a perfect world with no taxes, no brokerage costs, and infinitely divisible shares. You should also note that the MM discussion pertains to the firm's total *payout policy* (rather than to the narrower *dividend policy*).

Bird-in-hand (dividend preference theory) argument for dividend policy. When MM conclude that dividends are irrelevant, they mean that investors don't care about the firm's dividend policy since they can create their own. If they don't care, the firm's dividend policy will not affect the firm's stock price and, consequently, dividend policy will not affect the firm's required rate of return on equity capital (r_s). Myron Gordon and John Lintner, however, argue that r_s decreases as the dividend payout increases. Why? Because investors are less certain of receiving future capital gains from the reinvested retained earnings than they are of receiving current (and therefore certain) dividend payments. The main argument of Gordon and Lintner is that investors place a higher value on a dollar of dividends that they are certain to receive than on a dollar of *expected* capital gains. They base this argument on the fact that, when measuring total return, the dividend yield component, D_1 / P_0, has less risk than the growth component g.

PROFESSOR'S NOTE

The Gordon/Lintner argument is called the bird-in-the-hand-theory based on the old expression: a "bird in the hand" (dividends) is worth two in the bush (expected capital gains).

Tax aversion. In many countries, dividends have historically been taxed at higher rates than capital gains. In the 1970s, U.S. tax rates on dividend income were as high as 70%, while the taxes on capital gains were 35%. In the late 1990s, the rates were much lower, but the same general relationship was still in place: dividends were taxed as ordinary income with rates as high as 39.1%, while long-term capital gains were taxed at 20%. Under such a situation, according to the tax-aversion theory, investors will prefer to *not* receive dividends due to their higher tax rates. Taken to the extreme, the tax-aversion theory implies that investors would want companies to have a zero dividend payout ratio so that they will not be burdened with higher tax rates.

In the real world, tax laws often prevent companies from accumulating excess earnings, making dividend payments necessary. Also note that in 2003, tax laws in the United States changed so that dividends and long-term capital gains are both taxed at the same 15% rate.

Conclusions from the three theories. The results of empirical tests are unclear as to which of these theories best explains the empirical observations of dividend policy. Research suggests that higher tax rates do result in lower dividend payouts. In the United States, however, the change in tax law that put dividends and capital gains on common ground is likely to make the tax aversion theory irrelevant. There is empirical support for the "bird-in-the-hand" theory as some companies that pay dividends are perceived as less risky and specific groups of investors do prefer dividend paying stocks. MM counter this argument by saying that different dividend policies appeal to different clienteles, and that since all types of clients are active in the marketplace, dividend policy has no effect on company value if all clienteles are satisfied.

LOS 13.c: Describe types of information (signals) that dividend initiations, increases, decreases, and omissions may convey.

Information asymmetry refers to differences in information available to a company's board and management (insiders) as compared to the investors (outsiders). Dividends convey more credible information to the investors as compared to plain statements. This is so because dividends entail actual cash flow and are expected to be "sticky" (i.e., continue in the future). Companies refrain from increasing dividends unless they expect to continue to pay out the higher levels in the future. Similarly, companies loathe cutting dividends unless they expect that the lower levels of dividends reflect long-run poorer prospects of the company in the future.

The information conveyed by **dividend initiation** is ambiguous. On one hand, a dividend initiation could mean that a company is optimistic about the future and is sharing its wealth with stockholders—a positive signal. On the other hand, initiating a dividend could mean that a company has a lack of profitable reinvestment opportunities—a negative signal.

An **unexpected dividend increase** can signal to investors that a company's future business prospects are strong and that managers will share the success with shareholders. Studies have found that companies with a long history of dividend increases are dominant in their industries and have high returns on assets and low debt ratios.

Unexpected dividend decreases or **omissions** are typically negative signals that the business is in trouble and that management does not believe that the current dividend payment can be maintained. In rare instances, however, a dividend decrease or omission could be a positive sign. Management may believe that profitable investment opportunities are available and that shareholders would ultimately receive a greater benefit by having earnings reinvested in the company rather than being paid out as dividends.

LOS 13.d: Explain how agency costs may affect a company's payout policy.

Agency Costs

Between shareholders and managers: Agency costs reflect the inefficiencies due to divergence of interests between managers and stockholders. One aspect of agency issue is that managers may have an incentive to overinvest ("empire building"). This may lead to investment in some negative NPV projects, which reduces stockholder wealth. One way to reduce agency cost is to increase the payout of free cash flow as dividends. Generally, it makes sense for growing firms to retain a larger proportion of their earnings. However, mature firms in relatively non-cyclical industries do not need to hoard cash. In such cases, a higher dividend payout would be welcomed by the investors resulting in increases in stock value.

Between shareholders and bondholders: For firms financed by debt as well as equity, there may be an agency conflict between shareholders and bondholders. When there is risky debt outstanding, shareholders can pay themselves a large dividend, leaving the bondholders with a lower asset base as collateral. This way, there could be a transfer of wealth from bondholders to stockholders. Typically, agency conflict between stockholders and bondholders is resolved via provisions in the bond indenture. These provisions may include restrictions on dividend payment, maintenance of certain balance sheet ratios, and so on.

LOS 13.e: Explain factors that affect dividend policy in practice.

A company's dividend payout policy is the approach a company follows in determining the amount and timing of dividend payments to shareholders. Six primary factors affect a company's dividend payout policy:

1. **Investment opportunities.** Availability of positive NPV investment opportunities and the speed with which the firm must react to the opportunities determines the amount of cash the firm must keep on hand. If a firm faces many profitable investment opportunities and has to react quickly to capitalize on the opportunities (i.e., it does not have time to raise external capital), the dividend payout may be low.
2. **Expected volatility of future earnings.** Firms tie their target payout ratio to long-run sustainable earnings and are reluctant to increase dividends unless reversal is not expected in the near future. Hence, when earnings are volatile, firms are more cautious in *changing* dividend payout.
3. **Financial flexibility.** Firms with excess cash and a desire to maintain financial flexibility may resort to stock repurchases instead of dividends as a way to pay out excess cash. Since stock repurchase plans are not considered sticky (i.e., there is no implicit expectation by the market of an ongoing repurchase program), they don't entail reduction in financial flexibility going forward. Having cash on hand affords companies flexibility to meet unforeseen operating needs and investment opportunities. Financial flexibility is especially important during times of crisis when liquidity dries up and credit may be hard to obtain.

4. **Tax considerations.** Investors are concerned about after-tax returns. Investment income is taxed by most countries; however, the ways that dividends are taxed vary widely from country to country. The method and amount of tax applied to a dividend payment can have a significant impact on a firm's dividend policy. Generally, in countries where capital gains are taxed at a favorable rate compared to dividends, high-tax-bracket investors (like some individuals) prefer low dividend payouts, and low-tax-bracket investors (like corporations and pension funds) prefer high dividend payouts.

   A lower tax rate for dividends compared to capital gains does not necessarily mean companies will raise their dividend payouts. Stockholders may not prefer a higher dividend payout, even if the tax rate on dividends is more favorable, for multiple reasons:

   - Taxes on dividends are paid when the dividend is received, while capital gains taxes are paid only when shares are sold.
   - The cost basis of shares may receive a step-up in valuation at the shareholder's death. This means that taxes on capital gains may not have to be paid at all.
   - Tax-exempt institutions, such as pension funds and endowments, will be indifferent between dividends or capital gains.

5. **Flotation costs.** When a company issues new shares of common stock, a *flotation cost* of 3% to 7% is taken from the amount of capital raised to pay for investment bankers and other costs associated with issuing the new stock. Since retained earnings have no such fee, the cost of new equity capital is always higher than the cost of retained earnings. Larger companies typically have lower flotation costs as compared to smaller companies. Generally, the higher the flotation costs, the lower the dividend payout, given the need for equity capital in positive NPV projects.
6. **Contractual and legal restrictions.** Companies may be restricted from paying dividends either by legal requirements or by implicit restrictions caused by cash needs of the business. Common legal and contractual restrictions on dividend payments include:

   - The **impairment of capital rule.** A legal requirement in some countries mandates that dividends paid cannot be in excess of retained earnings.
   - **Debt covenants.** These are designed to protect bondholders and dictate things a company must or must not do. Many covenants require a firm to meet or exceed a certain target for liquidity ratios (e.g., current ratio) and coverage ratios (e.g., interest coverage ratio) before they can pay a dividend.

LOS 13.f: Calculate and interpret the effective tax rate on a given currency unit of corporate earnings under double taxation, dividend imputation, and split-rate tax systems.

Dividends paid in the United States are taxed according to what is called a **double-taxation system**. Earnings are taxed at the corporate level regardless of whether they are distributed as dividends, and dividends are taxed again at the shareholder level. In 2003, new tax legislation was passed in the United States that reduced the maximum tax rate on dividends at the individual shareholder level from 39.6% to 15%.

Since a dollar of earnings distributed as dividends is first taxed at the corporate level, with the after-corporate-tax amount taxed at the individual level, we can calculate the total effective tax rate as:

effective tax rate
= corporate tax rate + (1 − corporate tax rate)(individual tax rate)

EXAMPLE: Effective tax rate under a double taxation system

A U.S. company's annual earnings are $300, and the corporate tax rate is 35%. Assume that the company pays out 100% of its earnings as dividends. Calculate the effective tax rate on a dollar of corporate earnings paid out as dividends assuming a 15% tax rate on dividend income.

Answer:

Earnings	$300.00
(−) Tax @ 35%	(105.00)
Earnings after tax	195.00
Dividends (100% payout)	195.00
Tax on dividends (@ 15%)	(29.25)
After tax dividend to investor	165.75

$$\text{Effective (double) tax rate} = (300 - 165.75) / 300 = 44.75\%$$

or

$$0.35 + (1 - 0.35)(0.15) = 0.4475 \text{ or } 44.75\%.$$

A **split-rate** corporate tax system taxes earnings distributed as dividends at a lower rate than earnings that are retained. The effect is to offset the higher (double) tax rate applied to dividends at the individual level. Germany had a split-rate system until 2009. The calculation of the effective tax rate under a split-rate system is similar to the computation of the effective tax rate under double taxation except that the corporate tax rate applicable would be the corporate tax rate for distributed income.

EXAMPLE: Effective tax rate under a split-rate system

A German company's annual pretax earnings are €300. The corporate tax rate on retained earnings is 35%, and the corporate tax rate that applies to earnings paid out as dividends is 20%. Assuming that the company pays out 50% of its earnings as dividends, and the individual tax rate that applies to dividends is 30%, calculate the effective tax rate on one euro of corporate earnings paid out as a dividend.

Answer:

effective tax rate on income distributed as dividends
= 20% + [(1 − 20%) × 30%] = 44%

Note that under a split-rate system, earnings that are distributed as dividends are still taxed twice, but at a lower corporate tax rate (the corporate rate for distributed income).

Under an **imputation tax system**, taxes are paid at the corporate level but are attributed to the shareholder, so that *all taxes are effectively paid at the shareholder rate*. Shareholders deduct their portion of the taxes paid by the corporation from their tax return. If the shareholder tax bracket is lower than the company rate, the shareholder would receive a tax credit equal to the difference between the two rates. If the shareholder's tax bracket is higher than the company's rate, the shareholder pays the difference between the two rates.

EXAMPLE: Effective tax rate under an imputation system

Phil Cornelius and Ian Todd both own 100 shares of stock in a British corporation that makes £1.00 per share in pretax income. The corporation pays out all of its income as dividends. Cornelius is in the 20% individual tax bracket, while Todd is in the 40% individual tax bracket. The tax rate applicable to the corporation is 30%. Calculate the effective tax rate on the dividend for each shareholder.

Answer:

Effective Tax Rate Under an Imputation System

	Cornelius: 20% Rate	Todd: 40% Rate
Pretax income	£100	£100
Taxes at 30% corporate tax rate	£30	£30
Net income after tax	£70	£70
Dividend assuming 100% payout	£70	£70
Shareholder taxes on pretax income	£20	£40
Less tax credit for corporate payment	£30	£30
Tax due from shareholder	(£10)	£10
Effective tax rate on dividend	20 / 100 = 20%	40 / 100 = 40%

Under an imputation system, the effective tax rate on the dividend is simply the shareholder's marginal tax rate.

MODULE QUIZ 13.1

Use the following information to answer Questions 1 through 4.

Klaatu is a country that taxes dividends based on a double-taxation system. The corporate tax rate on company profits is 35%. Barada is a country that taxes dividends based on a split-rate tax system. The corporate tax rate applied to retained earnings is 36%, while the corporate tax rate applied to earnings paid out as dividends is 20%. Nikto is a country that taxes dividends based on an imputation tax system. The corporate tax rate on earnings is 38%.

1. An investor living in Klaatu holds 100 shares of stock in the Lucas Corporation. Lucas's pretax earnings for the current year are $2.00 per share, and the company has a payout ratio of 100%. The investor's individual tax rate on dividends is 30%. The effective tax rate on a dollar of funds to be paid out as dividends is *closest* to:

   A. 35.0%.
   B. 54.5%.

C. 62.3%.

2. An investor living in Barada holds 100 shares of Prowse, Inc. Prowse's pretax earnings in the current year are $1.00 per share, and Prowse pays dividends based on a target payout ratio of 40%. The individual tax rate that applies to dividends is 28%, and the individual tax rate that applies to capital gains is 15%. The effective tax rate on earnings distributed as dividends is:

   A. 20.0%.
   B. 42.4%.
   C. 53.9%.

3. Jenni White and Janet Langhals are each shareholders that live in Nikto, and each owns 100 shares of OCP, Inc., which has €1.00 per share in net income. OCP pays out 100% of its earnings as dividends. White is in the 25% tax bracket, while Langhals is in the 42% tax bracket. The effective tax rate on earnings paid out as dividends is:

   A. 28.0% for White and 42.0% for Langhals.
   B. 53.5% for White and 64.0% for Langhals.
   C. 25.0% for White and 42.0% for Langhals.

4. The bird-in-hand argument for dividend policy is based on the idea that:

   A. $r_s = D_1 / P_0 + g$ is constant for any dividend policy.
   B. a decrease in current dividends signals that future earnings will fall.
   C. because of perceived differences in risk, investors value a dollar of dividends more highly than a dollar of expected capital gains.

MODULE 13.2: STOCK BUYBACKS

Video covering this content is available online.

LOS 13.g: Compare stable dividend with constant dividend payout ratio, and calculate the dividend under each policy.

Stable Dividend Policy

The **stable dividend policy** focuses on a steady dividend payout, even though earnings may be volatile from year to year. Companies that use a stable dividend policy typically look at a forecast of their long-run earnings to determine the appropriate level for the stable dividend. This typically means aligning the company's dividend growth rate with the company's long-term earnings growth rate.

A stable dividend policy could be gradually moving towards a target dividend payout ratio. A model of gradual adjustment is called a **target payout adjustment model**.

Target Payout Adjustment Model

If company earnings are expected to increase and the current payout ratio is below the target payout ratio, an investor can estimate future dividends through the following formula:

expected increase in dividends = [(expected earnings × target payout ratio) – previous dividend] × adjustment factor

where:

adjustment factor = 1 / number of years over which the adjustment in dividends will take place

EXAMPLE: Expected dividend based on a target payout adjustment approach

Last year, Buckeye, Inc., had earnings of $3.50 per share and paid a dividend of $0.70. In the current year, the company expects to earn $4.50 per share. The company has a 35% target payout ratio and plans to bring its dividend up to the target payout ratio over a 5-year period. Calculate the expected dividend for the current year using the target payout adjustment model.

Answer:

expected increase in dividends

= [(expected earnings × target payout ratio) – previous dividend] × adjustment factor

= [($4.50 × 35%) – $0.70] × 0.2 = $0.175

expected dividend for the current year

= previous dividend + expected increase in dividends

= $0.70 + $0.175 = $0.875

PROFESSOR'S NOTE

Notice that the payout ratio actually *falls* from 20% to 19.4%. This is counterintuitive and seems to contradict the concept of the target payout ratio approach, which suggests that the firm should always be moving toward its target payout ratio (which in this case is 35%). Level II candidates are always troubled by the apparent inconsistency in the target payout adjustment approach. However, we can assure you that this example correctly applies the method and answers the LOS accurately. The target payout adjustment approach is expected to work in the long-run, even though year-to-year results may defy the logic behind it.

Constant Dividend Payout Ratio Policy

A payout ratio is the percentage of total earnings paid out as dividends. The constant payout ratio represents the proportion of earnings that a company plans to pay out to shareholders. A strict interpretation of the constant payout ratio method means that a company would pay out a specific percentage of its earnings each year as dividends, and the amount of those dividends would vary directly with earnings. This practice is seldom used.

LOS 13.h: Describe broad trends in corporate payout policies.

Dividend policy has been seen to have differences across countries and over time. This is probably to accommodate differences in investor preferences globally as well as changing investor preferences over time. The following generalizations can be made with respect to global trends in corporate payout policies:

1. Globally, in developed markets, the proportion of companies paying cash dividends has trended downwards over the long term.

2. The percentage of companies making stock repurchases has been trending upwards in the United States since the 1980s and in the United Kingdom and continental Europe since the 1990s. Major companies in Asia (particularly China and Japan) have made substantial repurchases since the 2010s.

LOS 13.i: Compare share repurchase methods.

Four common methods are used for share buybacks, though due to varying rules, some markets may not allow all methods. Outside the United States and Canada, method 1 is used almost exclusively.

1. **Open market transactions** are the most flexible approach, allowing a company to buy back its shares in the open market at the most favorable terms. There is no obligation on the part of the company to complete an announced buyback program. Unlike their European counterparts, American companies do not need shareholder approval for open market transactions.
2. **Fixed-price tender offer** is an approach where the firm buys a predetermined number of shares at a fixed price, typically at a premium over the current market price. Although the company forgoes flexibility (the firm cannot execute its purchases at an exact opportune time), it allows a company to buy back its shares rather quickly. If more than the desired number of shares are tendered in response to the offer, the company will typically buy back a prorated number of shares from each shareholder responding to the offer.
3. **Dutch auction** is a tender offer in which the company specifies not a single fixed price but rather a range of prices. Dutch auctions identify the minimum clearing price for the desired number of shares that need to be repurchased. Each participating shareholder indicates the price and the number of shares tendered. Bids are accepted based on lowest price first until the desired quantity is filled. The price of the last offer accepted (i.e., the highest accepted bid price) will however be the price paid for all shares tendered. Hence, a shareholder can increase the chance of having their tender accepted by offering shares at a low price. Dutch auctions also can be accomplished rather quickly, though not as quickly as tender offers.
4. **Repurchase by direct negotiation** entails purchasing shares from a major shareholder, often at a premium over market price. This method is often used in a greenmail scenario (where a hostile bidder is offered a premium to go away) to the detriment of the remaining shareholders. A negotiated purchase can also occur when a company wants to remove a large overhang in the market that is dampening the share price. Surprisingly, many negotiated transactions occur at a discount to market price, indicating that urgent liquidity needs of the seller motivated the transaction.

LOS 13.j: Calculate and compare the effect of a share repurchase on earnings per share when 1) the repurchase is financed with the company's surplus cash and 2) the company uses debt to finance the repurchase.

Repurchases made using a company's surplus cash will lower cash and shareholders' equity and, therefore, increase the firm's leverage (i.e., the debt-to-asset and debt-to-equity ratios). After the repurchase, earnings per share may increase (depending on the amount of cash used) because there will be fewer shares outstanding. If the repurchase was financed with additional debt offerings, the reduction in net income from the (after-tax) cost of the borrowed funds must also be factored in to determine the new impact on earnings per share.

EXAMPLE: Methods of financing a stock repurchase and their effect on EPS

JetFun, Inc., has 10 million shares outstanding and has just reported net income of $50 million. Because the company has $100 million of excess cash currently earning no return, JetFun's directors are considering repurchasing 2 million JetFun shares at a premium of 25% over the current market price of $40. Calculate and compare the effect of repurchase on EPS when the repurchase is (1) financed using the existing surplus cash, or (2) financed using new debt borrowed at an after-tax interest rate of 3%.

Answer:

Current EPS = $50 million / 10 million shares = $5

Repurchase price = $40 + 25% premium = 40(1.25) = $50 per share

Total cost of repurchase = $50 × 2 million shares = $100 million

Number of shares outstanding after buyback = 10 − 2 = 8 million shares

1. Using surplus cash:
   foregone income = $0
   new EPS = $50 million / 8 million shares = $6.25 (*an increase of 25% from $5*)
2. Using new debt:
   after-tax cost of funds = 3% of $100 million or $3 million
   earnings after deducting the cost of funds = 50 – 3 = $47 million
   new EPS = $47 million / 8 million shares = $5.875 (*an increase of 17.5% from $5*)

Note: Earnings yield before repurchase = EPS / price paid per share = 5/50 = 10%. When earnings yield is greater than the after-tax cost of debt (as is the case in both funding scenarios here), EPS will increase due to the repurchase.

From the previous example, we see that EPS will increase when the after-tax funding cost is less than the earnings yield. However, the firm will then also have higher leverage and, therefore, a higher cost of capital. Accordingly, an increase in EPS should not automatically lead to an increase in share price or in shareholder wealth.

LOS 13.k: Calculate the effect of a share repurchase on book value per share.

After a stock repurchase, the number of outstanding shares will decrease and the book value per share (BVPS) is likely to change as well. If the price paid is higher (lower) than the pre-repurchase BVPS, the BVPS will decrease (increase).

EXAMPLE: Impact of share repurchase on book value per share

Consider two companies, Alpha and Beta, which are both considering the repurchase of $5 million worth of shares. Additional details are given in the following:

	Alpha	Beta
Stock price	$20	$20
Number of shares outstanding	1 million	1 million
Book value	$15 million	$25 million

Calculate the impact of the repurchase transactions on BVPS.

Answer:

	Alpha	Beta
BVPS (before buyback)	15 / 1 = $15	25 / 1 = $25
Number of shares repurchased	$5 million / $20 = 250,000	$5 million / $20 = 250,000
Number of shares outstanding (after buyback)	1,000,000 – 250,000 = 750,000	1,000,000 – 250,000 = 750,000
Book value (after buyback)	15 – 5 = $10 million	25 – 5 = $20 million
BVPS (after buyback)	$10 million / 750,000 = $13.33	$20 million / 750,000 = $26.67

Alpha's BVPS of $15.00 was less than the $20.00 stock price, so after the buyback, Alpha's BVPS decreased to $13.33. Beta's BVPS increased from $25.00 to $26.67 after the buyback because the BVPS was greater than the $20.00 stock price.

LOS 13.l: Explain the choice between paying cash dividends and repurchasing shares.

There are five common rationales for share repurchases (versus dividends):

1. **Potential tax advantages.** When the tax rate on capital gains is lower than the tax rate on dividend income, share repurchases have a tax advantage over cash dividends.
2. **Share price support/signaling.** Companies may purchase their own stock, thereby signaling to the market that the company views its own stock as a good investment. Signaling is important in the presence of asymmetric information (where corporate insiders have access to better information about the company's prospects than outside investors). Management can send a signal to investors that the future outlook for the company is good. This tactic is often used when a share price is declining and management wants to convey confidence in the company's future to investors.
3. **Added flexibility.** A company could declare a regular cash dividend and periodically repurchase shares as a supplement to the dividend. Unlike dividends, share repurchases are not a long-term commitment. Since paying a cash dividend and repurchasing shares are economically equivalent, a company could declare a small stable dividend and then

repurchase shares with the company's leftover earnings. Additionally, managers have discretion with respect to "market timing" their repurchases.

4. **Offsetting dilution from employee stock options.** Repurchases offset EPS dilution that results from the exercise of employee stock options.
5. **Increasing financial leverage.** When funded by new debt, share repurchases increase leverage. Management can change the company's capital structure (and perhaps move toward the company's optimal capital structure) by decreasing the percentage of equity.

EXAMPLE: Impact of share repurchase and cash dividend of equal amounts

Spencer Pharmaceuticals, Inc., (SPI) has 20,000,000 shares outstanding with a current market value of \$50 per share. SPI made \$100 million in profits for the recent quarter, and since only 70% of these profits will be reinvested back into the company, SPI's Board of Directors is considering two alternatives for distributing the remaining 30% to shareholders:

- Pay a cash dividend of \$30,000,000 / 20,000,000 shares = \$1.50 per share.
- Repurchase \$30,000,000 worth of common stock.

Suppose that dividends are received when the shares go ex-dividend, the stock can be repurchased at the market price of \$50 per share, and there are no differences in tax treatment between the two alternatives. How would the wealth of an SPI shareholder be affected by the board's chosen method of distribution?

Answer:

(1) Cash dividend

After the shares go ex-dividend, a shareholder of a single share would have \$1.50 in cash and a share worth \$50 − \$1.50 = \$48.50.

The ex-dividend value of \$48.50 can also be calculated as the market value of equity after the distribution of the \$30 million, divided by the number of shares outstanding after the dividend payment.

$$\frac{(20{,}000{,}000)(\$50) - \$30{,}000{,}000}{20{,}000{,}000} = \$48.50$$

Total wealth from the ownership of one share = \$48.50 + \$1.50 = \$50.

(2) Share repurchase

With \$30,000,000, SPI could repurchase \$30,000,000 / \$50 = 600,000 shares of common stock. The share price after the repurchase is calculated as the market value of equity after the \$30,000,000 repurchase divided by the shares outstanding after the repurchase:

$$\frac{(20{,}000{,}000)(\$50) - \$30{,}000{,}000}{20{,}000{,}000 - 600{,}000} = \frac{\$970{,}000{,}000}{19{,}400{,}000} = \$50$$

Total wealth from the ownership of one share = $50.

This demonstrates that, assuming the tax treatment of the two alternatives is the same, a share repurchase has the same impact on shareholder wealth as a cash dividend payment of an equal amount.

In the previous example, we assumed that the company used cash to repurchase its stock. What if the company borrows funds to buy back the stock?

EXAMPLE: Share repurchase when the after-tax cost of debt is less than the earnings yield

Spencer Pharmaceuticals, Inc., (SPI) plans to borrow $30 million that it will use to repurchase shares. SPI's chief financial officer has compiled the following information:

- Share price at the time of buyback = $50.
- Shares outstanding before buyback = 20,000,000.
- EPS before buyback = $5.00.
- Earnings yield = $5.00 / $50 = 10%.
- After-tax cost of borrowing = 8%.
- Planned buyback = 600,000 shares.

Calculate the EPS after the buyback.

Answer:

$$\text{total earnings} = \$5.00 \times 20{,}000{,}000 = \$100{,}000{,}000$$

$$\begin{aligned}\text{EPS after buyback} &= \frac{\text{total earnings} - \text{after-tax cost of funds}}{\text{shares outstanding after buyback}} \\ &= \frac{\$100{,}000{,}000 - (600{,}000 \text{ shares} \times \$50 \times 0.08)}{(20{,}000{,}000 - 600{,}000) \text{ shares}} \\ &= \frac{\$100{,}000{,}000 - \$2{,}400{,}000}{19{,}400{,}000 \text{ shares}} \\ &= \frac{\$97{,}600{,}000}{19{,}400{,}000 \text{ shares}} \\ &= \$5.03 \text{ per share}\end{aligned}$$

Since the after-tax cost of borrowing of 8% is less than the 10% earnings yield (E/P) of the shares, the share repurchase will increase the company's EPS.

EXAMPLE: Share repurchase with borrowed funds, where the after-tax cost of debt *exceeds* the earnings yield

Spencer Pharmaceuticals, Inc., (SPI) plans to borrow $30 million that it will use to repurchase shares; however, creditors perceive the company to be a significant credit risk, and the after-tax cost of borrowing has jumped to 15%. Using the other information from the previous example, calculate the EPS after the buyback.

Answer:

$$\text{EPS after buyback} = \frac{\text{total earnings} - \text{after-tax cost of funds}}{\text{shares outstanding after buyback}}$$

$$= \frac{\$100{,}000{,}000 - (600{,}000 \text{ shares} \times \$50 \times 0.15)}{(20{,}000{,}000 - 600{,}000) \text{ shares}}$$

$$= \frac{\$100{,}000{,}000 - \$4{,}500{,}000}{19{,}400{,}000 \text{ shares}}$$

$$= \frac{\$95{,}500{,}000}{19{,}400{,}000 \text{ shares}}$$

$$= \$4.92 \text{ per share}$$

Because the after-tax cost of borrowing of 15% exceeds the earnings yield of 10%, the added interest paid reduces earnings, and the EPS after the buyback is less than the original $5.00.

The conclusion is that a share repurchase using borrowed funds will increase EPS if the after-tax cost of debt used to buy back shares is less than the earnings yield of the shares before the repurchase. It will decrease EPS if the cost of debt is greater than the earnings yield, and it will not change EPS if the two are equal.

LOS 13.m: Calculate and interpret dividend coverage ratios based on 1) net income and 2) free cash flow.

LOS 13.n: Identify characteristics of companies that may not be able to sustain their cash dividend.

Dividend safety is the metric used to evaluate the probability of dividends continuing at the current rate for a company. Traditional ratios, such as **dividend payout ratio** (dividends/net income) or its inverse **dividend coverage ratio** (net income/dividends), are typically used for this purpose. A higher than normal dividend payout ratio (and lower than normal dividend coverage ratio) tends to typically indicate a higher probability of a dividend cut (or a lower probability of dividend sustainability).

In analyzing these ratios, we should compare the computed ratio to the average ratio for the industry and market within which a company operates. In making a qualitative judgment about a company, stable or increasing dividends are looked upon favorably, whereas companies that have cut their dividends in the past are looked upon unfavorably.

Another ratio considers free cash flow to equity (FCFE). FCFE is the cash flow available for distribution to stockholders after working capital and fixed capital needs are accounted for.

FCFE coverage ratio = FCFE / (dividends + share repurchases)

Note that unlike the dividend payout ratio, the FCFE coverage ratio considers not only dividends but also share repurchases.

A FCFE coverage ratio significantly less than one is considered unsustainable. In such a situation, the company is drawing down its cash reserves for dividends and repurchases.

PROFESSOR'S NOTE

FCFE is discussed extensively in the Equity Valuation portion of the curriculum.

EXAMPLE: Dividend sustainability analysis

Chevron Corp. (NYSE: CVX) is a San Ramon, CA based global integrated energy company. Selected financial data for the years ending December 31, 2008 and 2009 is provided.

Year Ending 31 December (US $ Millions)	2009	2008
Net income	$10,483	$23,931
Cash flow from operations	$19,373	$29,632
Capital expenditures (FCInv)	$19,843	$19,666
Net borrowing	$1,659	$1,682
Dividends paid	$5,373	$5,261
Stock repurchases	$(168)	$6,821

Source: Yahoo! Finance website. July 27, 2010.

1. Using the information provided, calculate:
   A. Dividend payout ratio.
   B. Dividend coverage ratio.
   C. FCFE coverage ratio.
2. Discuss the trend in dividend coverage ratio and compare it to FCFE coverage ratio.
3. Is Chevron's dividend sustainable?

Answer:

1. a. Dividend payout ratio = dividend / net income.
   2008: 5,261 / 23,931 = 0.22 or 22%
   2009: 5,373 / 10,483 = 0.51 or 51%

   b. Dividend coverage ratio = net income / dividend
   2008: 23,931 / 5,261 = 4.55
   2009: 10,483 / 5,373 = 1.95

   c. FCFE = cash flow from operations − FCInv + net borrowings
   2008: FCFE = 29,632 – 19,666 + 1,682 = 11,648
   2009: FCFE = 19,373 – 19,843 + 1,659 = 1,189
   FCFE coverage ratio = FCFE / (dividends + share repurchases)
   FCFE coverage ratio:
   2008: 11,648 / (5,261 + 6,821) = 0.96
   2009: 1,189 / [5,373 + (–168)] = 0.23

2. The dividend coverage ratio has decreased considerably from 4.55 in 2008 to 1.95 in 2009. The FCFE coverage ratio has also decreased significantly from 0.96 to 0.23. It appears that Chevron's dividend sustainability is lower in 2009 compared to 2008.
3. Chevron's dividend coverage has decreased significantly from 2008 to 2009. Still, even with this decrease, Chevron's dividend coverage is almost 2 times. Chevron's FCFE

coverage ratio has also decreased dramatically from 2008 to 2009. This is despite the fact that the company stopped its share repurchase plan in 2009 (in fact it issued additional stock—most likely to provide shares for exercise of employee stock options). Additionally, even if we consider 2009 to be a bad year due to a downturn in oil prices, the 2008 ratio is still below one. FCFE coverage ratio analysis therefore suggests that Chevron's payout policy is unsustainable for the long run.

MODULE QUIZ 13.2

1. Over the past 25 years, in the developed markets including the United States, the United Kingdom, and the European Union, the fraction of companies that:
   A. pay out cash dividends has decreased.
   B. repurchase shares has decreased.
   C. use particular dividend policies has been consistent over time and across countries.
2. Which of the following is *most likely* to be sustainable?
   A. A FCFE coverage ratio of 0.5.
   B. A dividend payout ratio of 0.5.
   C. A dividend coverage ratio of 0.5.
3. Nick Adams is recommending to the board of directors that they share the profits from an excellent year (totaling $56 million) with shareholders by either declaring a special cash dividend of $20 million, or using the $20 million to repurchase shares of Volksberger common stock in the open market. Selected financial information about the firm is shown in the following:

Shares outstanding:	40 million
Current stock price:	$28.00
52-week trading range:	$20.00 to $36.00
Book value of equity:	$880 million
After-tax cost of borrowing:	5.5%

   Adams drafts a memo to the board of directors detailing the financial impact of declaring a special cash dividend versus repurchasing shares. His memo includes the following two statements:

   Statement 1: The total shareholder wealth resulting from owning one share of stock with the special dividend option will increase to $28.50.

   Statement 2: Our company's P/E ratio after the share buyback will remain the same as before the buyback.

   Which of Adams's statements are correct?
   A. Both statements are correct.
   B. Only one of the statements is correct.
   C. Neither statement is correct.
4. Which of the following would not be a good reason for a company to repurchase shares of its own stock? Management:
   A. believes a stable cash dividend is in the best interests of shareholders.
   B. believes its stock is overvalued.
   C. wants to increase the amount of leverage in its capital structure.
5. Last year, Wolverine Shoes and Boots had earnings of $4.00 per share and paid a dividend of $0.20. In the current year, the company expects to earn $4.40 per share. The company has a

30% target payout ratio and plans to bring its dividend up to the target payout ratio over an 8-year period. Next year's expected dividend is *closest* to:

A. $0.14.
B. $0.24.
C. $0.34.

KEY CONCEPTS

LOS 13.a

Cash dividend payments reduce cash as well as stockholders' equity. This results in a lower quick ratio and current ratio, and higher leverage (e.g., debt-to-equity and debt-to-asset) ratios. Stock dividends (and stock splits) leave a company's capital structure unchanged and do not affect any of these ratios. In the case of a stock dividend, a decrease in retained earnings (corresponding to the value of the stock dividend) is offset by an increase in contributed capital, leaving the value of total equity unchanged.

LOS 13.b

Following are the three theories of investor preference:

- MM's dividend irrelevance theory holds that in a no-tax/no-fees world, dividend policy is irrelevant since it has no effect on the price of a firm's stock or its cost of capital, because individual investors can create their own homemade dividend.
- Dividend preference theory says investors prefer the certainty of current cash to future capital gains.
- Tax aversion theory states that investors are tax averse to dividends and would prefer companies instead buy back shares, especially when the tax rate on dividends is higher than the tax rate on capital gains.

LOS 13.c

The signaling effect of dividend changes is based on the idea that dividends convey information about future earnings from management to investors (who have less information about a firm's prospects than management). In general, unexpected increases are good news and unexpected decreases are bad news as seen by U.S. investors.

LOS 13.d

Two types of agency costs affect dividend payout policies:

- Agency conflict between shareholders and managers can be reduced by paying out a higher proportion of the firm's free cash flow to equity so as to discourage investment in negative NPV projects.
- Agency conflict between shareholders and bondholders occurs when shareholders can expropriate bondholder wealth by paying themselves a large dividend (and leaving a lower asset base for outstanding bonds as collateral). Agency conflict between bondholders and stockholders is typically resolved via provisions in bond indenture.

LOS 13.e

Six primary factors affect a company's dividend payout policy:

1. Investment opportunities: affects the residual income available to pay as dividends.
2. Expected volatility of future earnings: firms are more cautious in changing dividend payout in the presence of high earnings volatility.
3. Financial flexibility: Firms may not increase dividends (even in presence of significant free cash flow) so as not to be forced to continue paying those dividends in the future and losing financial flexibility. Instead, firms can choose to pay out excess cash via stock repurchases.
4. Tax considerations: In the presence of differential tax rate on capital gains versus dividends, companies may structure their dividend policy to maximize investors' after-tax income.
5. Flotation costs: Flotation cost increases the cost of external equity as compared to retained earnings. Hence, higher flotation cost would motivate firms to have a lower dividend payout.
6. Contractual and legal restrictions: Dividend policy may be affected by debt covenants that the firm has to adhere to. Legal restrictions in some jurisdictions limit the dividend payout of a firm.

LOS 13.f

Effective rate under double taxation
= corporate tax rate + (1 − corporate tax rate) × (individual tax rate)

A split-rate system has different corporate tax rates on retained earnings and earnings that are paid out in dividends (distributed income). Under split-rate system, effective tax rate is computed the same way as double taxation but we use the corporate tax rate for distributed income as the relevant corporate tax rate in the double taxation formula.

Under a tax imputation system, taxes are paid at the corporate level but are used as credits by the stockholders. Hence, all taxes are effectively paid at the shareholder's marginal tax rate.

LOS 13.g

Stable dividend policy: A company tries to align its dividend growth rate with the company's long-term earnings growth rate to provide a steady dividend. A firm with a stable dividend policy could use a target payout adjustment model to gradually move towards its target payout.

$$\text{expected increase in dividends} = [(\text{expected earnings} \times \text{target payout ratio}) - \text{previous dividend}] \times \text{adjustment factor}$$

where:

$$\text{adjustment factor} = 1 / \text{number of years over which the adjustment in dividends will take place}$$

Constant payout ratio: Company defines a proportion of earnings that it plans to pay out to shareholders regardless of volatility in earnings (and consequently in dividends).

LOS 13.h

Global trends in corporate payout policies:

1. Globally, the proportion of companies paying cash dividends has trended downwards.

2. Stock repurchases have been trending upwards in the United States since the 1980s and in the United Kingdom and continental Europe since the 1990s.

LOS 13.i

Share repurchase methods:

1. Open market transactions: The firm buys back its shares in the open market.
2. Fixed-price tender offer: The firm buys a predetermined number of shares at a fixed price, typically at a premium over the current market price.
3. Dutch auction: A tender offer where the company specifies a range of prices rather than a fixed price. Bids are accepted (lowest price first) until the desired quantity is filled. All accepted bids are then filled at the (higher) price of the last accepted bid.
4. Repurchase by direct negotiation: Purchasing shares from a major shareholder, often at a premium over market price. This method may be used in a greenmail scenario, or when a company wants to remove a large overhang in the market.

LOS 13.j

Repurchases made using a company’s surplus cash will lower cash and shareholders’ equity and, therefore, increase the firm’s leverage. Earnings per share may increase because there will be fewer shares outstanding.

When the company uses debt to finance the repurchase, EPS will increase if the after-tax funding cost is less than the earnings yield. However, the firm will then also have higher leverage and, therefore, a higher cost of capital, so an increase in EPS will not automatically lead to an increase in share price or in shareholder wealth.

LOS 13.k

After a stock repurchase, the number of outstanding shares will decrease and the book value per share (BVPS) is likely to change as well. If the price paid is higher (lower) than the pre-repurchase BVPS, the BVPS will decrease (increase).

LOS 13.l

There are five common rationales for share repurchases (versus dividends):

1. Potential tax advantages: When capital gains are taxed favorably as compared to dividends.
2. Share price support/signaling: Management wants to signal better prospects for the firm.
3. Added flexibility: Reduces the need for “sticky” dividends in the future.
4. Offsets dilution from employee stock options.
5. Increases financial leverage by reducing equity in the balance sheet.

LOS 13.m

Dividend coverage ratio = net income / dividends

FCFE coverage ratio = FCFE / (dividends + share repurchases)

LOS 13.n

For both dividend and FCFE coverage, ratios that are below industry averages or trending downwards over time indicate problems for dividend sustainability.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 13.1

1. **B** The effective tax rate on earnings distributed as dividends is 0.35 + (1 − 0.35)(0.30) = 0.545 = 54.5%. (LOS 13.f)

2. **B** The effective tax rate on earnings distributed as dividends is 0.20 + (1 − 0.20)(0.28) = 0.424 = 42.4%. (LOS 13.f)

3. **C** Under an imputation tax system, the effective tax rate on earnings distributed as dividends is the tax rate of the shareholder receiving the dividends. (LOS 13.f)

4. **C** The bird-in-hand argument for dividend policy is based on the fact that a dividend payment is more certain than future capital gains. (LOS 13.b)

Module Quiz 13.2

1. **A** Over the past decades, it has been observed that the percentage of companies engaging in share repurchases has increased over time. At the same time, the fraction of companies paying cash dividends has decreased. In addition to changing over time, dividend policies have been noted to differ between countries. (LOS 13.h)

2. **B** An FCFE coverage ratio or dividend coverage ratio much less than one is not sustainable because the company is drawing on cash and marketable securities to make payments. A dividend payout ratio less than one indicates that the company is paying out less in dividends than it is earning, which is the normal (and desirable) situation. (LOS 13.n)

3. **C** Adams is incorrect with respect to Statement 1. If the firm pays its special dividend of $20 million, both the assets and equity of the firm will drop by $20 million. The total wealth from owning one share will be [(40 million)($28) − $20 million] / 40 million = $27.50, plus $20 million / 40 million = $0.50 per share as a dividend, so the total shareholder wealth resulting from owning one share of stock is $28. Note that the total shareholder wealth of $28 is the same whether the cash dividend or share repurchase option is chosen. Adams is also incorrect with respect to Statement 2. The current EPS is $56 million / 40 million = $1.40, so the current P/E ratio is $28 / $1.40 = 20 times earnings. The price per share will remain the same. Share buyback = $20 million / 28 = 714,286 shares. New price = [(40 million × $28/share) − $20 million] / (40 million − 714,286) = $28/share. EPS will increase. $56 million / 39,285,714 = $1.43. Since the price is the same, and EPS increases, the P/E ratio will fall slightly after the repurchase. (LOS 13.j)

4. **B** Management would repurchase shares of its own stock if it believed the shares were undervalued, not overvalued. (LOS 13.l)

5. **C** expected increase in dividends

= [(expected earnings × target payout ratio) – previous dividend] × adjustment factor

= [($4.40 × 30%) – $0.20] × (1 / 8)

= $0.14

expected dividend for the current year

= previous dividend + expected increase in dividends

= $0.20 + $0.14

= $0.34

(LOS 13.g)

READING 14

ENVIRONMENTAL, SOCIAL, AND GOVERNANCE (ESG) CONSIDERATIONS IN INVESTMENT ANALYSIS

EXAM FOCUS

In this topic review, we consider how environmental, social, and governance (ESG) factors affect analysis of a company. First, we think about global variations in corporate ownership structures and how corporate governance is impacted by these structures. Next, we reflect on various company-specific corporate governance factors. We then identify the various ESG risks and opportunities to analyze. Finally, we consider some examples that demonstrate how ESG risk and opportunities are evaluated.

INTRODUCTION

In the field of investment analysis, factors related to ESG have become an increasingly important consideration. For example, environmental risks have taken on a greater focus as the public devotes more thought to conservation of the planet's resources and as corresponding regulations are strengthened globally. In general, a firm today should give considerable attention to its impact on the local community, its employee health policies, and how the company promotes itself.

MODULE 14.1: GLOBAL VARIATIONS IN OWNERSHIP STRUCTURES

Video covering this content is available online.

LOS 14.a: Describe global variations in ownership structures and the possible effects of these variations on corporate governance policies and practices.

Global differences in legal, social, political, and economic factors have led to differences in ownership structures.

Concentrated Ownership vs. Dispersed Ownership

Corporate ownership structures can be classified as either concentrated, dispersed, or a hybrid. Globally, the most common ownership structure category is concentrated ownership, where a single shareholder or a group of shareholders have control over the corporation. These controlling shareholders can be a family, other companies, or a sovereign entity. Dispersed ownership refers to the situation where the shareholders are numerous and none has control.

Percentage ownership is not always a reliable indicator of concentration of control: certain shareholders may have a greater degree of control than their ownership percentage would suggest. A minority shareholder (i.e., holding less than 50% of shares) could have control of a corporation through a **vertical ownership** arrangement (where the group has a controlling interest in holding companies, which in turn have controlling interests in operating companies) or a **horizontal ownership** arrangement (where companies with common suppliers or customers cross-hold each other's shares).

Another disconnect between control and percentage ownership comes in the form of dual-class shares, wherein one class of shareholders has fewer voting rights, while the other class has superior voting rights.

Conflicts Within Different Ownership Structures

Depending on the type of ownership structure present in a corporation, different kinds of conflicts may emerge among shareholders and between managers and shareholders.

Dispersed ownership and dispersed voting power describes a situation where shareholders (called *weak shareholders*) do not hold power over managers (called *strong managers*). In this situation, principal–agent conflict is likely: shareholders want shareholder value maximized, while managers may use the firm's resources to their own advantage. This problem can be mitigated by the presence of controlling shareholders.

Concentrated ownership and concentrated voting power refers to a situation where so-called "strong" shareholders hold power over minority shareholders and "weak" managers. This arrangement allows controlling shareholders to control the board of directors and effectively control and monitor management. The downside of this situation is that a principal–principal problem may arise: controlling owners can take advantage of firm resources to the detriment of minority owners.

Dispersed ownership and concentrated voting power indicates that the majority of the shares of a company are not owned by strong controlling shareholders. Rather, the controlling shareholders gain control over other minority shareholders through pyramid structures or dual-class shares, despite the controlling shareholders having less-than-majority ownership (again, the principal–principal problem). These controlling shareholders' voting power also allows them to monitor management.

Concentrated ownership and dispersed voting power occurs in the presence of voting caps, where the voting rights of large share positions are restricted. Sovereign countries sometimes enact voting caps to discourage foreign investors from taking a controlling position in a company belonging to an industry that is considered important.

Categories of Influential Shareholders

To properly identify various corporate governance risks, it can be helpful to consider the various types of shareholders and how they may impact corporate governance.

1. **Banks.** When a bank lends money to a firm or holds an equity stake, the bank can often exert some control over that corporation. (This is especially common in Asia and Europe.) If a bank is both a lender and a shareholder to a firm, the bank may use its influence to encourage the firm to take out large expensive loans from the bank. Corporate governance controls here could ensure that the bank does not take advantage of its role as lender at the expense of other shareholders.
2. **Families.** In Latin America and some other places, family ownership is common. It is also common for individuals to sit on the boards of numerous companies, a term called *interlocking directorates*, resulting in one family controlling multiple companies. One advantage of family control is that principal–agent issues may be reduced. On the other hand, family ownership can make it difficult to recruit quality outsiders for management and often leads to lack of concern for minority shareholders, as well as minimal transparency and low accountability by management.
3. **State-owned enterprises.** When a corporation operates in a sector that the government feels is strategically important, or needs to produce some good or product essential to a country's population, or if the capital required exceeds what the private sector can fund, the corporation may be operated as a state-owned enterprise (SOE). A **listed SOE** is partly owned by the government and also trades on an exchange. It may be the case that SOEs will seek to provide social benefits to the public rather than focusing only on shareholder value maximization.
4. **Institutional investors.** In many countries, institutional investors (e.g., pension funds, insurance companies, mutual funds) can represent a large portion of equity ownership, and tend to wield their shareholder rights with expertise because of the experience and resources that they bring to the table. While institutional investors usually will not own a large enough position to be a controlling shareholder, they can typically pressure a firm's management and board to act in the interest of shareholders.
5. **Group companies**, such as South Korea's Samsung, can achieve an outsized amount of control through cross-holding of shares via vertical and horizontal ownership. These cross-holding schemes make it difficult for outsiders to acquire shares. Cross-holding also increases the potential for the firms to participate in related-party transactions that do not advantage minority shareholders.
6. **Private equity firms** invest in private companies (or public companies that they plan to take private). The involvement of private equity firms may bring beneficial changes to the portfolio firms' corporate governance, such as introducing performance-based compensation for managers, or the addition of codes.
7. **Foreign investors.** Foreign investors typically demand greater accountability and transparency. This is particularly true in emerging market countries. Minority shareholders can benefit when a company decides to cross-list its shares in a country that has a greater protection for investors and higher standards of transparency.
8. **Managers and board directors.** When the board of directors and company managers own shares in a firm, their interests become more aligned with those of other shareholders, and these insiders are more likely to use the firm's resources to boost profitability over the long term. On the other hand, there is also the potential for insiders to use their ownership stakes to protect their own interests rather than those of other shareholders.

Ownership Structure and Corporate Governance

Some of the major ownership structure factors that impact corporate governance include:

1. **Director independence.** When a board member has no significant remuneration, ownership, or employment relationship with the firm, the board member is considered independent. Independent directors are important in countries with dispersed ownership where the principal–agent problem is greater and thus the board's role of monitoring managers is key. The portion of directors on boards that are independent has grown in the aftermath of corporate scandals (e.g., Enron).
2. **Board structures.** Boards of directors can generally be categorized as one- or two-tier. A one-tier board is the most common and is made up of both internal (executive) directors and external (nonexecutive) directors. Under a two-tier structure (required in countries such as Russia, Germany, and Argentina), the **management board** is overseen by a **supervisory board**. This supervisory board performs functions such as determining management compensation, supervising external auditors, and reviewing the firm's financial records. In some jurisdictions, representatives of stakeholders such as labor groups sit on the supervisory board.
3. **Special voting arrangements.** Some countries attempt to provide an advantage to minority shareholders through special voting arrangements for board nomination and election.
4. **Corporate governance codes, laws, and listing requirements.** Some countries have adopted national "comply or explain" corporate governance codes that require firms to either adopt best practices of corporate governance or explain why they have not.
5. **Stewardship codes.** Stewardship codes exist in some countries that seek to engage investors in corporate governance by exercising their legal rights. In some cases, these codes are voluntary, but in the U.K., for example, institutional investors are required to "comply or explain" with respect to the U.K.'s Stewardship Code.

MODULE QUIZ 14.1

1. The principal–agent problem is *most severe* under a corporate ownership structure of:
   A. dispersed ownership and dispersed voting power.
   B. concentrated ownership and concentrated voting power.
   C. dispersed ownership and concentrated voting power.
2. A corporation's two-tier board of directors is *most likely* to consist of a(n):
   A. executive board that oversees an internal board.
   B. external board that oversees a nonexecutive board.
   C. supervisory board that oversees a management board.

MODULE 14.2: EVALUATING ESG EXPOSURES

Video covering this content is available online.

LOS 14.b: Evaluate the effectiveness of a company's corporate governance policies and practices.

It is important for a firm's corporate governance to be effective. Consequences of ineffective corporate governance can include damage to reputation, difficulty competing, low or volatile stock price, higher borrowing costs, and decreased profitability. When a company is not behaving according to stockholders' wishes, **shareholder activism** may occur: this is the term for techniques used by shareholders to force management to act in shareholders' interests.

Board Policies and Practices

A board's policies and practices determine its effectiveness.

1. **Structure of board of directors.** When analyzing the board structure, the analyst should focus on whether the board is appropriate in light of its accountability to shareholders and its oversight and representation. **CEO duality** occurs when the chairperson of the board is also the chief executive officer (CEO). CEO duality raises concerns that the chairperson's oversight and monitoring responsibilities may not be effectively rendered.
2. **Board independence.** Ideally, a majority of board members should be independent; independent directors are more likely to prevent management from self-serving behavior. It will also lower investor perceptions of risk.
3. **Board committees.** The number and types of board committees will vary greatly between companies but often include committees related to governance issues such as compensation, nomination, and audit committees, as well as in areas such as risk and compliance. When analyzing a corporation's board, an analyst should consider whether the key committees related to financial reporting, management selection, and compensation are sufficiently independent.
4. **Skills and experience of board.** Ideally, board members will have industry-specific experience and skills while also having the broad expertise needed to function efficiently in the board member role. When the company faces specific ESG risks, it is essential that board members have some previous exposure to those kinds of concerns.

   Board tenure is another consideration. A board member's long tenure (say, more than 10 years) can be viewed as either an asset or a liability. A long-tenured board member will have strong knowledge of how the firm's management has been operating, as well as a good understanding of the firm's business operations. However, long tenure can make a board member resistant to beneficial change and can cause board members to become too friendly with management, affecting the board members' independence.

5. **Composition of board.** It has been observed in recent years that small, diverse boards of directors are generally more effective than large boards made up of members with similar backgrounds. Diversity refers to characteristics such as age, gender, length of tenure, education, culture, and place of birth.
6. **Other board evaluation considerations.** A board of directors may be evaluated from a number of perspectives, but an evaluation is generally undertaken to appease investors and to ensure that the firm is operating effectively. Typical areas of interest include the board's structure and committees, the culture of the board, its interaction with management, its effectiveness, and its leadership. Board evaluations can be self-evaluations or external reviews (by outsiders). The audience for a particular evaluation can be investors, regulators, or some other interested party.

Executive Compensation

When the remuneration of a firm's executives is analyzed, one of the primary concerns is whether this compensation scheme properly aligns management's incentives with the corporation's value creation. Metrics such as KPIs (key performance indicators) are useful measures of incentive mechanisms. **Clawback policies** allow firms to reclaim past compensation if inappropriate conduct comes to light later (e.g., violations of the law, misreported financial statements). **Say-on-pay** rules can give stakeholders the opportunity to vote on executive compensation. The pay differential between the CEO and the firm's average worker has come under scrutiny in recent years.

Voting Rights of Shareholders

When investing in a company, it is important to be aware of the rights associated with holding various classes of shares. Some firms structure share offerings as a single class with equal voting rights. Alternatively, in a dual-class structure, the shares held by the firm's founders or management have more voting power than those sold to external investors. Such a structure will disadvantage minority shareholders and sets up a conflict of interest between the two classes of shareholders.

LOS 14.c: Describe how ESG-related risk exposures and investment opportunities may be identified and evaluated.

As awareness of ESG-related risks has increased over time among stakeholders, the amount of related disclosures has also risen. However, obtaining sufficient quality and quantity of information about ESG concerns is still a challenge for investment analysts. Sources of ESG data can include corporate communications as well as corporate filings. However, there is no uniformity in such disclosures, and they are usually purely voluntary.

Investment Horizon and Materiality

Analysts should consider the impact of identified ESG factors over time; issues that are likely to have a financial impact on the firm only over the long term may be of little concern for short-term investors. Further, the analyst should determine if the impact is material or not. **Materiality** refers to an impact on the company's share price, its operations, or other financial aspects. In general, an investor would consider an ESG factor material if it would affect their investment decisions.

Relevant ESG Factors

Before an analyst starts compiling data related to a specific company's ESG exposures, the analyst should first work to determine which specific ESG factors are most relevant to that particular industry and firm. Several methods might be used to identify these factors, including ESG data providers, industry organizations, and proprietary methods.

1. **ESG data providers.** Data is sourced from a vendor of ESG information, such as MSCI, Sustainalytics, or RepRisk. These vendors provide information in the form of rankings, scores, and quantitative analysis.

2. **Industry organizations.** These not-for-profit groups can provide information about ESG factors. Examples include the IIRC (International Integrated Reporting Council) and the SASB (Sustainable Accounting Standards Board). These organizations work to promote standardized corporate disclosures of ESG issues. For example, the SASB seeks to promote consistent accounting standards for sustainability.
3. **Proprietary methods.** The analyst uses her firm's tools along with her judgment to gather ESG data from published reports, government organizations, and other sources. ESG data specific to a particular firm can be derived from sources such as 10-K regulatory filings, corporate sustainability reports, and annual reports. While considerations like the board structure may be similar across many companies, ESG factors are likely to vary significantly between industries. For example, a financial institution may have exposures related to issues of data security, while an energy company is more likely to be exposed to environmental concerns. While various organizations are working toward consistent standards relating to reporting ESG factors, comparability continues to be a challenge for analysts.

Security Analysis: Fixed Income vs. Equity

ESG analysis is applicable both to fixed-income and equity securities, and the methods used to identify applicable ESG factors are essentially the same. However, when it comes to valuation of securities, there are differences between the valuation of fixed-income and equity instruments in terms of how ESG factors are used.

Fixed-Income Analysis

Fixed-income analysts usually will focus on ESG factors' downside risk. For example, an analyst evaluating a company that could be subject to lawsuits might make corresponding adjustments to that firm's liquidity ratios, cash flow ratios, credit ratios, or credit spreads. The timing of such a lawsuit should be considered: the impact of a potential lawsuit far in the future on short-maturity instruments is likely to be minimal. Similarly, for a company with equipment that is likely to become obsolete (known as a **stranded assets** issue), it is the value of the firm's 20-year bonds that should be particularly sensitive to the risk that the equipment will no longer be viable in 20 years.

Equity Analysis

Equity analysts consider both the upside and downside impact of ESG factors when valuing a firm's stock. For example, the analyst may use a lower discount rate for a company that has implemented new manufacturing practices that are environmentally friendly compared to those of competitors. Conversely, the analyst may increase the projected operating expenses of a firm that is struggling with employee retention.

Green bonds are fixed-income instruments used to fund projects related to the environment. Aside from the intended use of the proceeds, these bonds tend to have the same recourse and credit ratings as the issuer's other bonds. Valuation of these bonds is similar to that of a conventional bond, except that green bonds may command a price premium. One concern is greenwashing, where a bond's proceeds are not actually used for the environmental purpose advertised.

LOS 14.d: Evaluate ESG risk exposures and investment opportunities related to a company.

ESG considerations are increasingly being taken into account during the investment process. In the following sections, we will consider some examples of how ESG factors can be included in a fixed-income or equity valuation.

ESG Integration

ESG factors are generally included in a firm's valuation by making various adjustments to the company's financial statements. For example, a company's projected cash flow statement or income statement may be adjusted in terms of earnings, margins, revenues, costs, capital expenditures, or other items. Adjustments to the balance sheet often take the form of altering the value of assets to reflect impairment. The credit spread of a fixed-income instrument may be adjusted to reflect ESG concerns. Similarly, an adjustment to discount rate or cost of capital may be used to reflect ESG considerations in an equity analysis.

ESG Integration Examples

We illustrate the ESG analytical principles involved using three fictitious companies.

1. **Environmental factors.** Concerned about wasting water, a soft-drink company has been able to lower its water usage in its manufacturing process over the past three years. The corresponding analyst adjustment is to reduce projected cost of goods sold (as a percentage of revenues). This leads to improved gross margin, higher forecasted earnings, and a higher stock price. This may also lead to a modest improvement in CDS or bond prices, due to slightly improved creditworthiness.
2. **Social factors.** A drug company has experienced a long string of product recalls, product quality controversies, and fines and warning letters from regulators. To reflect that these negative factors may recur in the future, the analyst makes some adjustments: COGS (a as percentage of revenues) are increased, and projected non-operating expenses (as a percentage of operating income) related to restructuring are increased as well. Negative valuation impacts for stock and bondholders are also appropriate, especially if the firm's brand reputation is impaired.
3. **Governance factors.** Compared to its peers, a bank's board is found to be lagging in a number of measures: the bank's chairperson is not independent, the board's overall independence and diversity is low, board members' industry experience is low, and a number of board members have long tenures. In addition to these factors, the bank has a higher ratio of nonperforming loans than its peers. To reflect these additional risks, the analyst uses a higher risk premium to value the bank's stock and also increases the credit spread used to value the bank's debt.

MODULE QUIZ 14.2

1. *CEO duality* refers to a situation where the chief executive officer (CEO):
   A. is a shareholder in the company.
   B. serves as chairperson of the board.

C. has been granted superior or even sole voting rights.

2. In fixed-income analysis, ESG integration is:

   A. generally focused on identifying downside risk.
   B. used to identify both potential opportunities and downside risk.
   C. generally focused on identifying potential opportunities.

KEY CONCEPTS

LOS 14.a

Shareholder ownership is usually categorized as concentrated, dispersed, or a hybrid.

Dispersed ownership indicates that none of the many shareholders has control over the corporation.

Concentrated corporate ownership means that controlling shareholders (i.e., an individual or group) can exercise control over the company. The controlling shareholders can be either minority or majority shareholders.

Vertical (also known as pyramid) ownership refers to a structure where a company has a controlling interest in multiple holding companies, and those holding companies hold controlling interests in operating companies. Horizontal ownership refers to companies with shared business interests that cross-hold the shares of each other.

Dual-class shares give superior (or sole) voting rights to one share class and lesser (or no) voting rights to another.

The board of directors of a company can be structured either as a one-tier board consisting of internal (executive) and external (non-executive) directors, or a two-tiered board where the management board is overseen by a supervisory board.

LOS 14.b

The situation where the CEO is also the company's chairperson of the board is called CEO duality. CEO duality raises the concern that, relative to having independent chairperson and CEO roles, oversight and monitoring roles of the board could be compromised.

LOS 14.c

ESG information and metrics are inconsistently reported by companies, and such disclosure is voluntary. This makes it difficult for analysts to identify useful and relevant ESG factor data.

Materiality in ESG refers to an issue that could impact a firm's securities, performance, or operations.

There are three main approaches for identifying a company's ESG factors: 1) ESG data providers, 2) industry organizations, and 3) proprietary methods.

In fixed-income analysis, ESG considerations are primarily concerned with downside risk. In equity analysis, ESG is considered both in regards to upside opportunities and downside risk.

LOS 14.d

Analysts evaluate ESG factors and then make corresponding adjustments to estimate a discount rate or risk premium. ESG factor adjustments related to income statement and statement of cash flows relate to projected revenues, costs, margins, earnings, capex, or other line items. ESG adjustments to a firm's balance sheet often involve evaluating potential impairment of the firm's assets.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 14.1

1. **A** Under the combination of dispersed ownership and dispersed voting power, the principal–agent problem is likely to arise: this is the risk that managers will use a company's resources to pursue their own interests. The other two ownership structures mentioned are more closely associated with the principal–principal problem. (LOS 14.a)

2. **C** Two-tier boards of directors consist of a supervisory board that oversees a management board. A one-tier board is a single board of directors that is composed of internal (i.e., executive) and external (i.e., nonexecutive) directors. (LOS 14.a)

Module Quiz 14.2

1. **B** CEO duality refers to the situation when the chief executive officer also serves as chairperson of the board. When a board director or manager is also a shareholder of a company, they are known as insiders. Dual-class shares generally grant one or more share classes superior or even sole voting rights. (LOS 14.b)

2. **A** In fixed-income analysis, ESG integration is generally focused on identifying downside risk. (In equity analysis, ESG integration is used to identify both potential opportunities and downside risk.) (LOS 14.c)

READING 15

COST OF CAPITAL: ADVANCED TOPICS

EXAM FOCUS

This reading focuses on estimation of companies' cost of capital. Understand the qualitative factors that influence the risk premiums on debt and equity, and how equity risk premium is calculated based on historical data or on forward-looking estimates. Finally, understand the complications involved in estimating the cost of capital for privately held and international companies.

WARM-UP: WEIGHTED AVERAGE COST OF CAPITAL

The **weighted average cost of capital (WACC)** for a company is calculated as:

$$WACC = W_e r_e + W_p r_p + W_d r_d(1 - T)$$

where:

r_e = cost of equity

r_p = cost of preferred equity

r_d = cost of debt

and W_e, W_p, and W_d are the respective weights.

We consider the *after-tax* cost of debt if the interest expense is tax-deductible (using the marginal tax rate for the company).

PROFESSOR'S NOTE

The weights in the equation just listed should be *target* weights of the company. Unless told otherwise, assume that current market value-based weights reflect the company's target capital structure.

MODULE 15.1: FACTORS AFFECTING THE COST OF CAPITAL AND THE COST OF DEBT

LOS 15.a: Explain top-down and bottom-up factors that impact the cost of capital.

Top-down (i.e., external or macro) factors include:

- *Capital availability*. Economies with plentiful availability of capital will have lower cost of capital. Developed economies with established liquid capital markets tend to have more stable currency values and better investor protections via rule of law, and accordingly will have greater capital availability compared to developing economies. In some less-developed countries that lack corporate debt markets, businesses have to rely on bank lending or a **shadow banking** system of unregulated, non-bank sources.
- *Market conditions*. Factors include inflation, interest rates, and the state of the economy. Lower levels of expected inflation lead to lower nominal risk-free rates. Risk premiums on both debt and equity shrink during economic expansions, and rise during economic contractions. The transparent and predictable monetary policies of developed economies tend to reduce risk premiums and interest rates. Finally, countries with higher currency volatility will need to offer higher risk premiums to risk-averse investors.
- *Legal and regulatory considerations*, *country risk*. Countries that follow common law–based legal systems offer stronger protection to investors, leading to lower risk premiums compared to countries with civil law–based legal systems.
- *Tax jurisdiction*. Tax code affects the deductibility of interest expense on debt. All else equal, the higher the marginal tax rate, the greater the tax benefit of using debt in the capital structure.

Bottom-up (i.e., company-specific) factors that affect the cost of capital include:

- *Business or operating risk*. Volatility in revenues, earnings, and cash flows is a measure of business risk. Businesses with stable revenues (e.g., utilities or subscription-based services) have relatively stable earnings and cash flows, and are therefore less risky than businesses with more-volatile revenues (e.g., cyclical industries). Companies generating most of their revenues from only a few customers face **customer concentration risk**. Use of debt in the capital structure increases financial leverage, which increases the volatility of earnings and cash flows. Companies with poor corporate governance, as well as companies with high ESG-related risk exposures, will see investors demand higher risk premiums.
- *Asset nature and liquidity*. Company assets form the collateral for servicing debt in the event of a liquidation. Companies with tangible, non-specialized (i.e., fungible) assets and more liquid assets would have a higher recovery rate and hence a lower risk premium. Specialized assets and intangibles (e.g., goodwill, patents, proprietary production processes, etc.) do not have a ready liquid market, resulting in a lower recovery rate. Assets specifically designated as collateral reduce the cost of secured debt, but increase the cost of other subordinated unsecured debt as their claim becomes inferior.
- *Financial strength and profitability*. Companies with higher profitability, higher ability to generate cash, and lower leverage, have a lower probability of default; therefore, investors will accept a lower risk premium. Leverage ratios such as debt-to-equity or debt-to-EBITDA are used by analysts.
- *Security features*. Embedded call options make a security less desirable and increase the risk premium. While a callable bond increases the current cost of borrowing for the issuer, it does allow the company to refinance the debt at a favorable rate should interest rates decline in the future. Conversely, a put option or conversion option

reduces the cost of borrowing for the company, because a putable bond is issued at a favorable rate. However, the borrowing cost may increase in the future if investors put back the bond in a rising interest rate scenario and the company is forced to refinance at a higher rate. A **cumulative preferred stock** accumulates missed dividends when the company is unprofitable; such stock has a lower risk premium than equivalent non-cumulative preferred stock. Classes of common equity that have superior rights have a lower cost than those classes with inferior rights.

LOS 15.b: Compare methods used to estimate the cost of debt.

LOS 15.c: Estimate the cost of debt or required return on equity for a public company and a private company.

Estimated cost of debt is impacted by various factors, including whether the debt is publicly traded, whether it is rated by a ratings agency, and which currency the debt was issued in.

Publicly Traded Debt

If the company's debt is publicly traded, the yield to maturity for the longest maturity straight debt outstanding is generally the best estimate of the cost of debt. However, if the longest maturity debt is thinly traded, and a shorter-maturity debt with more-reliable market price information is available, we may use the yield on this shorter maturity debt instead.

Non-Traded/Thinly Traded

If the company's debt is not traded or is thinly traded, we can use matrix pricing to consider the yields on traded securities with the same maturity and credit ratings. If the debt is not credit rated, then financial ratios of the company such as the **interest coverage (IC) ratio** or financial leverage (D/E) ratio may be used to infer a credit rating for that debt. It is also common to obtain credit spreads for specific ratings and add those to the benchmark rates to arrive at the cost of debt. When using credit ratings, analysts should realize that an issuer rating may differ from the ratings of the issuer's individual series of debt, depending on the seniority of the specific series and whether it is secured by a collateral.

EXAMPLE: Cost of debt using matrix pricing

Brevis Solutions is a technology provider for the healthcare sector. Sunil Tilak, CFA is trying to estimate the cost of debt, which represents 30% of Brevis' capital structure. The 6-year, BB rated debt is thinly traded. Tilak collects data, shown here, on similar BB rated securities that have liquid markets.

Company	Maturity (yrs)	Annual Coupon	Price (Par = $100)
Silva	4	5%	$ 99.45
Deso	4	6%	$ 101.75
Manfried	7	7%	$ 110.00
Listor	7	8%	$ 114.00

Estimate the cost of debt using the matrix method.

Answer:

First calculate the YTM of each of the four comparable securities.

Company	Maturity (yrs)	Annual Coupon	Price (Par = $100)	YTM
Silva	4	5%	$ 99.45	5.16%
Deso	4	6%	$ 101.75	5.50%
Manfried	7	7%	$ 110.00	5.26%
Listor	7	8%	$ 114.00	5.53%

Using your calculator, for Listor: PV = −114, N = 7, PMT = 8, FV = 100, CPT I/Y = 5.53%

We then construct a matrix of maturity and coupon rates as follows:

	Coupon Rate				
Maturity	5%	6%	7%	8%	Average
4	5.16%	5.50%	–	–	5.33%
7	–	–	5.26%	5.53%	5.39%

The average yields for each maturity are then calculated and used as estimates for those maturities.

Finally, we use linear interpolation to determine the YTM for a 6-year bond as follows:

interpolated yield = $\text{yield}_{\text{short}} + [(\text{yield}_{\text{long}} - \text{yield}_{\text{short}})/(\text{maturity}_{\text{long}} - \text{maturity}_{\text{short}})] \times (\text{maturity}_{\text{interpolated}} - \text{maturity}_{\text{short}})$

= 5.33% + [(5.39 − 5.33) / (7 − 4)] × (6 − 4)

= 5.33% + (0.02% × 2)

= 5.37%

In many countries, bank debt is the primary source of debt capital. Because we are interested in the marginal cost or the cost of a new bank loan, the rate on existing debt will not be a good estimate if the company's characteristics or the general level of rates has changed.

Leases—and specifically, finance (i.e., capital) leases—can be used to estimate the cost of borrowing. The **rate implicit in the lease (RIIL)** is the implied cost of capital in a lease. RIIL is the IRR in the following equation:

PV of lease payments + PV of residual value = Fair value of leased asset + Lessor's initial direct cost

Usually, the terms of a finance lease are disclosed. If they are not, then the **incremental borrowing rate (IBR)**, the rate on a new secured loan over the same term, may be used.

EXAMPLE: Capital lease

Company A has signed a 15-year lease on an asset, calling for annual payments of $10 million at the end of each year. The lease residual value is $30 million. The fair value of the asset is $120 million, and the lessor incurs a cost of $5 million at the time of lease initiation.

Calculate the RIIL for this lease.

Answer:

Using our financial calculator:

PV = −125, N = 15, PMT = 10, and FV = 30. CPT I/Y = 4.28%

Thus, the rate implicit in the lease (RIIL) is 4.28%.

International Considerations

For foreign borrowers, the cost of debt should include a country risk premium. A **country risk rating (CRR)** reflects risks related to economic conditions, political stability, exchange rate risk, and the level of capital market development. Country ratings can be similar in format to debt ratings (e.g., AA), or can be a numerical rating (e.g., 8).

A country may be assigned a rating relative to a benchmark country. The excess of the median interest rate for that country relative to the benchmark country rate determines the country risk premium. In Figure 15.1, Country A is the benchmark country, and hence that country's risk premium is zero. Country C's median interest rate is 6.80% resulting in a country risk premium of 6.80% − 4.20% = 2.60%.

Figure 15.1: Country Risk Premium

Country	CRR (100 = Least Risky)	Median Interest Rate	Country Risk Premium
A	100	4.20%	0
B	90	5.60%	1.40%
C	80	6.80%	2.60%
D	70	7.20%	3.00%
E	60	8.25%	4.05%

MODULE 15.2: ERP AND THE COST OF EQUITY

LOS 15.c: Explain historical and forward-looking approaches to estimating an equity risk premium.

Equity risk premium (ERP) is the extra return (over the risk-free rate) that an investor demands for investing in risky equity securities.

Estimates of the Equity Risk Premium

There are two types of estimates of the equity risk premium: historical estimates and forward-looking estimates.

Historical Estimates

A historical estimate of the ERP consists of the difference between the historical mean return for a broad-based equity market index over some time period, and a risk-free rate over the same period. Its strength is its objectivity and simplicity.

Analysts making a historical estimate need to decide on four important issues:

1. Index selection: The equity market index chosen should be one that serves as a good proxy for the average returns equity investors earn over time.
2. Sample period: Analysts generally choose a longer sample period because covering multiple business cycles and a variety of market conditions, the standard error (i.e., noise) should be lower. However, older data may not reflect current market conditions. On the other hand, smaller samples using only recent data can have a higher forecast error but may be representative of expected market conditions.
3. Mean type: Analysts need to decide whether to use an arithmetic mean or a geometric mean. An arithmetic mean is a good estimate of a one-period expected return, but does a poor job of estimating multiperiod return, which is needed to determine expected terminal wealth. A geometric mean gives lower weight to outliers, and estimates the expected terminal wealth more accurately. While the geometric mean is preferred, both means are used in practice.
4. Risk-free rate proxy: Analysts may use either short-term or long-term government bond rates as a proxy for the risk-free rate. The short-term (e.g., T-bill) rate is a good proxy for the true risk-free rate because (unlike long-term rates) it does not include reinvestment (of coupon) risk. However, because the duration of equity is long-term (actually infinite), many analysts favor using long-term government bond rates as proxies for the risk-free rate.

A weakness of the historical approach is the assumption that the mean and variance of the returns are constant over time (i.e., the ERP time series is stationary). This does not seem to be the case. In fact, the equity risk premium actually appears to be countercyclical—it is low during good times and high during bad times. Thus, an analyst using historical data to estimate the current equity premium should choose the sample period carefully. Another concern with a historical estimate is that it may suffer from **survivorship bias**: it will be upward-biased if only firms that have survived during the period of measurement are included in the sample.

Forward-Looking Estimates

Forward-looking or ex-ante estimates use current information and expectations concerning economic and financial variables. As such, they do not rely on an assumption of stationarity and is less subject to problems like survivorship bias.

There are three main categories of forward-looking estimates: survey-based estimates, dividend discount model estimates, and estimates from macroeconomic modeling.

Survey Estimates

Survey estimates of the ERP use a consensus of opinions, typically experts. Surveys tend to estimate higher ERPs for developing markets relative to developed markets. A weakness is that the surveys tend to be biased toward recent market returns.

Dividend Discount Models

The constant growth model (Gordon Growth Model) is a popular method for generating forward-looking estimates. Rearranging the GGM to isolate the required rate of return gives:

$$r_e = (D_1/V_0) + g$$

The first term is the expected dividend yield on a broad-based index; this measure should be relatively free from large surprises. The second term is the earnings growth rate for the index, based on consensus forecasts. Recall that the constant growth rate is also the capital gains yield on the index.

Subtracting the current risk-free rate gives us the ERP estimate:

$$\text{ERP} = \text{E(dividend yield)} + g - r_f$$

Recall that the constant growth model assumes a constant growth rate in earnings and dividends. If this assumption is not valid, an analyst will need to include relevant expectations of expansion or contraction of the market P/E ratio. (We will discuss this in a moment in the macroeconomic model approach).

For developing economies, where the current high growth rate is transitory, we might incorporate three stages of growth (high, moderate, and mature) and model the stock market index value this way:

current index value = PV(stage 1) + PV(stage 2) + PV(stage 3)

The IRR that balances this equation is then our estimate of required rate of return. After subtracting the current risk-free rate, we can obtain the ERP for that market.

Macroeconomic Models

We will use the Grinold-Kroner (2001) model to illustrate a macroeconomic model. The **Grinold-Kroner model** decomposes ‘g’ (the capital gains yield, CGY) in the dividend discount model as:

$$\text{CGY} = \Delta P/E + \text{expected inflation } (i) + \text{real economic growth rate (G)} - \Delta S$$

where:

$\Delta P/E$ = expected repricing or expansion/contraction in P/Es

ΔS = % change in shares outstanding, on an aggregate basis

ΔS reflects the net buyback of stock, which is another way corporations can effectively pay cash to their shareholders. If ΔS is positive, it is a dilution effect (hence, is subtracted).

The Grinold-Kroner model can be used to express the expected market equity return as a function of five factors:

$$\text{ERP} = [\text{DY} + \Delta\text{P/E} + i + \text{G} - \Delta\text{S}] - r_f$$

where:

DY = dividend yield; the expected income component of the equity investment

i = expected inflation

One approach to estimate expected inflation is to compare the nominal yield on a U.S. Treasury bond to the yield on an equivalent inflation-protected Treasury security (TIPS).

$$\text{i} = \frac{1 + \text{YTM}_{\text{Treas}}}{1 + \text{YTM}_{\text{TIPS}}} - 1$$

EXAMPLE: Macroeconomic model

Patrick McGill is trying to estimate the equity risk premium for the U.S. market. McGill uses the S&P 500 index as the proxy for the market and estimates that the dividend yield is 1.2%. The real GDP growth rate is forecast to be 3% and McGill believes that the market is currently fairly valued. The current 10-year Treasury yield is 2.4%, while 10-year TIPS yield 0.25%. McGill assumes no net change in shares outstanding, and that the risk-free rate is 0.50%.

Calculate the equity risk premium.

Answer:

$$\text{i} = \frac{1 + \text{YTM}_{\text{Treas}}}{1 + \text{YTM}_{\text{TIPS}}} - 1 = (1.024 \,/\, 1.0025) - 1 = 2.1\%$$

$$\begin{aligned}\text{ERP} &= [\text{DY} + \Delta\text{P/E} + i + \text{G} - \Delta\text{S}] - r_f \\ &= [1.2\% + 0 + 2.1\% + 3\% - 0] - 0.5\% \\ &= 5.8\%\end{aligned}$$

LOS 15.d: Compare methods used to estimate the required return on equity.

LOS 15.e: Estimate the cost of debt or required return on equity for a public company and a private company.

There are several approaches to estimating the **required rate of return on equity** including the DDM, bond yield plus risk premium, build-up approaches, and risk premium models.

Dividend Discount Model (DDM)

For an individual company, cost of equity is dividend yield plus capital gains yield.

r_e = DY + CGY

(Note that since we are calculating required return and not risk premium, *we do not subtract the risk-free rate*.)

If the company's earnings growth rate was not constant, then a 2-stage model could be used instead.

EXAMPLE: Cost of equity using DDM

Cogenics, Inc., is expected to pay a dividend of $4 at the end of its first year. Dividends are expected to grow at a constant rate of 4% per year. The current Cogenics stock price is $100.

Betagenics, Inc., is expected to pay a dividend of $1.50, $2.00, $2.50, and $3.00 at the end of each of the next four years, respectively. The current Betagenics stock price is $50, and is expected to be $60 at the end of four years.

Based on this data, calculate the cost of equity for Cogenics and Betagenics.

Answer:

Dividend yield for Cogenics = 4 / 100 = 4%

$r_{e(Cogenics)}$ = DY + CGY = 4% + 4% = 8%

For Betagenics, we need to solve for r_e in the following equation:

$$\$50 = \frac{1.50}{(1+r_e)} + \frac{2.00}{(1+r_e)^2} + \frac{2.50}{(1+r_e)^3} + \frac{3.00+60}{(1+r_e)^4}$$

On our TI BA II PLUS calculator: CF0 = −50; C01 = 1.50; C02 = 2.00; C03 = 2.50; C04 = 63

CPT IRR = 8.78%

Thus, the cost of equity for Betagenics is 8.78%.

Bond-Yield-Plus–Risk-Premium Method

The **bond-yield-plus-risk-premium (BYRPM) method** is a build-up approach appropriate for estimating the cost of equity for a company that has publicly traded debt. This method simply adds a risk premium to the yield to maturity (YTM) of the company's *long-term* debt. One approach to selecting a risk premium is to use an average of the historical difference between equity returns and cost of debt, similar to the historical estimation of the equity risk premium.

An issue with this method is that the risk premium is rather arbitrary and if the firm has several debt securities outstanding, there is no perfect option regarding which security's yield to use.

EXAMPLE: Applying the bond-yield-plus-risk-premium approach

Company LMN has bonds with 15 years to maturity, a coupon of 8.2%, and a price of 101.70. An analyst estimates that the additional risk assumed from holding the firm's equity rather than bonds justifies a risk premium of 3.8%. Calculate the cost of equity using the bond-yield-plus-risk-premium approach.

Answer:

Using our calculator's TVM keys: PV = −101.70, N = 15, PMT = 8.2, FV = 100, CPT I/Y = 8.0%.

Cost of equity = 8.0% + 3.8% = 11.8%

Risk-Based Models

Capital Asset Pricing Model

The **capital asset pricing model (CAPM)** is a single-factor model that estimates the required return on equity using the following formula:

required return on stock = risk-free rate + (equity risk premium × beta of stock)

EXAMPLE: Using the CAPM to calculate the required return on equity

The expected risk-free rate is 4%, and the equity risk premium is 3.9%. Calculate the required return on equity for a stock with beta of 0.8.

Answer:

required return on stock

= risk-free rate + (equity risk premium × beta of stock)

= 4% + (3.9% × 0.8) = 7.12%

CAPM beta can be estimated using the **market model**, where we regress the historical return on the stock against the corresponding return on a broad-based equity index. Note that the returns used are generally excess returns, that is, return over the risk-free rate, but raw returns can also be used. The estimated slope coefficient in this simple linear regression is the estimated beta of the stock.

To estimate the beta of a private company, the beta of a comparable publicly traded company may be used (after adjusting for leverage differences).

Multifactor Models

Multifactor models can have greater explanatory power than the CAPM (which is a single-factor model).

The general form of an *n*-factor multifactor model is:

required return = r_f + (risk premium)$_1$ + (risk premium)$_2$ + … + (risk premium)$_n$

where:

(risk premium)$_i$ = (factor sensitivity)$_i$ × (factor risk premium)$_i$

The factor sensitivity (also called the *factor beta*) is the asset's sensitivity to a particular factor, all else equal. The factor risk premium is the expected return above the risk-free rate from a unit (i.e., = 1) sensitivity to the factor (and zero sensitivity to all other factors).

Fama–French Models

The **Fama–French model** is a multifactor model that adds two additional factors to the CAPM systematic risk factor: size (measured by market cap) and value (measured by the book-to-market ratio).

required return of stock = $r_f + \beta_1\text{ERP} + \beta_2\text{SMB} + \beta_3\text{HML}$

where:

SMB = size premium = average difference in return of small-cap portfolios over large-cap portfolios

HML = value premium = average difference in return of high book-to-market portfolios over low book-to-market portfolios.

A five-factor Fama–French model adds to the model two additional factors: profitability (RMW), and investment factor (CMA).

required return of stock = $r_f + \beta_1\ \text{ERP} + \beta_2\ \text{SMB} + \beta_3\ \text{HML} + \beta_4\ \text{RMW} + \beta_5\ \text{CMA}$

where:

RMW = profitability premium = average difference in portfolio returns of companies with "robust profitability" over "weak profitability"

CMA = investment premium = average difference in portfolio returns of companies with "conservative" investments over companies with "aggressive" investments

EXAMPLE: **Cost of equity using the Fama–French Five-Factor Model**

Suppose the current risk-free rate of return is 2.1%. Calculate the cost of equity for Fulton Corp. given the following information.

Factor	Beta	Risk Premium
Market	1.1	3.2%
Size	0.2	1.3%
Value	–0.3	2.0%
Profitability	0.18	4.2%
Investment Style	0.5	2.4%

Answer:

r_e = 2.1% + (1.1 × 3.2%) + (0.2 × 1.3%) + (–0.3 × 2%) + (0.18 × 4.2%) + (0.5 × 2.4%) = 7.2%

Cost of Equity for Private Companies

Private companies are often smaller and less mature than public companies. Since they have no market price data, the CAPM and Fama–French models are not suitable to apply directly to private companies. Illiquidity and the lack of transparency (e.g., no security filings and disclosures) increase the investment risk of private companies.

Appropriate risk premiums for a private company include: **size premium (SP)**, **industry risk premium (IP)**, and **specific company risk premium (SCRP)**. SCRP is a general risk premium covering unique risks of the private company such as key-person risk, geographical location risk, etc., which are difficult to diversify away.

Factors Affecting SCRP

Qualitative factors: Industry in which the company operates, corporate governance quality, asset nature and type (e.g., tangible, liquid, fungible, etc.), customer and supplier concentration, geographical concentration, competitive position in the industry, and management quality. Lower corporate governance quality, large proportion of intangibles in total assets, poor competitive standing in the industry, management without adequate skill and experience, geographical concentration, customer concentration, or supplier concentration all indicate higher risk.

Quantitative factors: Operating and financial leverage, earnings volatility and predictability, and cash flow volatility. Higher leverage and volatility indicate higher risk.

There are two approaches to estimate the cost of equity for private companies—expanded CAPM and build-up approach.

Expanded CAPM

The **expanded CAPM** approach starts with using the standard CAPM (with a peer beta) and then adding risk premiums as needed:

required return = $r_f + (\beta_{peer} \times ERP) + SP + SCRP$

where:

β_{Peer} = the industry beta from peer public companies

Build-Up

The **build-up approach** starts with the risk-free rate, and adds the equity risk premium to arrive at the required return on an *average large* public company (because ERP is calculated using a market-cap weighted index, large caps dominate the index, and no beta is used). Further risk premiums for size, industry, and company-specific characteristics are added as needed, depending on how different the subject company from the average large public company.

required return = r_f + ERP + SP + IP + SCRP

International Considerations

While CAPM may work well for developed market securities, risks unique to emerging markets require additional premiums. The **country-spread model** and the **country risk rating model** are alternative ways to estimate these risk premiums.

The country-spread model estimates a **country risk premium (CRP)** (also called a country spread premium) for a specific emerging market. The estimated equity risk premium then becomes:

$ERP_{\text{emerging market}} = ERP_{\text{developed}} + (\lambda \times CRP)$

where:

λ = exposure of the company to the local (emerging market) economy.

We can use the **sovereign yield spread** (the difference in yields of emerging market government securities relative to developed market benchmark security) as an estimate of CRP. However, differences in legal and market environment complicate the use of just yield spreads for CRP.

Damodaran recommended adjusting the sovereign yield spread by the ratio of standard deviation of the country's equity and bond markets:

CRP = sovereign yield spread × (σ_{equity} / σ_{bond})

Extended CAPM

For countries operating globally, several approaches are used to estimate r_e including global CAPM, international CAPM, and country spread and risk rating models.

Global CAPM (GCAPM) uses a global market index to estimate the ERP, rather than using only a local market index. Because of low correlation between developed markets and emerging markets, the beta coefficient using a global market proxy is usually quite low (or even negative). To overcome this, a second factor representing the local market is sometimes included; but the availability of reliable market index data is a concern in many emerging markets.

International CAPM (ICAPM) is a 2-factor model, based on (1) a global market index (e.g., MSCI All Country World Index) factor, and (2) a foreign currency–denominated, wealth-weighted market index.

$$E(r_e) = r_f + \beta_G[E(r_{gm}) - r_f] + \beta_C[E(r_C) - r_f]$$

where:

β_G = sensitivity to the global market index
r_{gm} = global market return
β_C = sensitivity to the foreign currency index
r_C = foreign currency index return

The first factor captures the company's relationship with the local economy relative to the global economy: lower values of β_G indicate lower integration of the company with the global economy. The second factor captures the sensitivity of the company's cash flows to changes in its local currency exchange rate.

In summary, for companies with global operations that are limited to developed markets, GCAPM or ICAPM are appropriate approaches to estimating the required return on equity. For companies with exposure to developing markets, the appropriate approach is less well-defined. For these firms, a country risk premium (CRP) can be used, if we can assume that the historical estimate is representative of the risk premium going forward.

LOS 15.f: Evaluate a company's capital structure and cost of capital relative to peers.

This LOS uses a case study approach to pull together the concepts learned in the prior LOS. Exam questions need to be solvable in approximately three minutes each, so a full-fledged case study on exam day is impractical. However, individual questions can be fair game, as illustrated in the module quiz.

MODULE QUIZ 15.1, 15.2

Use the following information to answer Questions 1 through 6.

Daniel Hsu, CFA, follows the pharmaceutical industry for Applied Fundamentals LLC. Hsu has asked associate Rudolph Hsei to evaluate the cost of capital in two countries P and Q. Hsei collects the information shown in Exhibit 1.

Exhibit 1: Selected Information for Countries P and Q

Factor	Country P	Country Q
Legal System	Common law	Civil law
Capital Markets	Developed	Less developed
Volatility of currency value	Low	High

Hsu is interested in estimating the cost of capital for two companies: Sizemore and Minicure. Sizemore is a large-cap company with a rich portfolio of successful drugs, while Minicure is a privately held company with a single drug, Mitra, which was approved by the FDA for treatment of early-stage dementia.

Exhibit 2 shows selected financial information relevant to Sizemore.

Exhibit 2: Sizemore, Selected Data ($ in millions)

Net revenues	17,942
EBITDA	4,498
EBIT	3,421
Interest expense	814
Income tax	662
Net income	1,945
Total debt	8,671
Cash and cash equivalents	2,400
Stockholders' equity	17,342

Sizemore's tax rate is 34%.

After analyzing hundreds of rated securities, Hsu arrives at a matrix for interest coverage (IC) ratios and financial leverage (D/E) ratios by rating class and credit spread relative to the benchmark security (currently yielding 2.3%) as shown in Exhibit 3.

Exhibit 3: Rating Classes, Ratios and Spreads (truncated)

Rating Class	IC	D/E	Spread
AAA	IC > 10	D/E < 25%	0.25%
AA	7 < IC < 10	25% < D/E < 35%	0.50%
A	5 < IC < 7	35% < D/E < 40%	0.80%
BBB	4 < IC < 5	40% < D/E < 52%	1.20%
BB	3 < IC < 4	52% < D/E < 60%	1.75%
B	2 < IC < 3	60% < D/E < 67%	2.33%
CCC	1 < IC < 2	67% < D/E < 75%	3.20%

While discussing the computation of the equity risk premium, Hsei makes the following statements:

Statement 1: The arithmetic mean of historical values of the equity risk premium (ERP) provides the best estimate of terminal value of wealth if invested at the ERP.

Statement 2: Forward-looking estimates of the equity risk premium (ERP) do not suffer from survivorship bias and do not rely on the ERP being stationary.

Hsu obtains estimates of economic data for use in the calculation of ERP as shown in Exhibit 4.

Exhibit 4: Selected Data: Wilshire 5000

Expected dividend yield	1.10%
Forecast P/E growth rate	−0.10%
Forecast real GDP growth	3%
10-year Treasury yield	2.67%
10-year TIPS yield	0.33%
Change in shares outstanding	0

Hsu then compiles consensus estimates of Sizemore's dividends and stock price. Sizemore is expected to pay a dividend of $1.00 per share in each of the next three years, and the stock price is expected to be $34 at the end of three years. Sizemore's current stock price is $25.

Hsei collects the following information and ponders how they might affect the SCRP for Minicure:

Factor 1:	Minicure's founder is the chairman and CEO. The management team is comprised of the team of scientists whose research initially led to the development of Mitra.
Factor 2:	The value of the patent on Mitra represents 90% of Minicure's total assets.
Factor 3:	Subcontractor Samita, Inc., handles the commercial production of Mitra because Samita is the only company that has the necessary specialized equipment.

1. Based on the information in Exhibit 1, Hsei should conclude that relative to Country P, Country Q's cost of capital would be *most likely* be:
   A. higher.
   B. lower.
   C. approximately the same.
2. Based on information provided in Exhibits 2 and 3 and elsewhere, the after-tax cost of debt for Sizemore should be *closest* to:
   A. 2.31%.
   B. 3.50%.
   C. 4.05%.
3. Which statements made by Hsei regarding the equity risk premium are correct?
   A. Statement 1 only.
   B. Statement 2 only.
   C. Neither statement is correct.
4. Using information in Exhibit 4, the estimate of equity risk premium (ERP) consistent with the Grinold-Kroner model is *closest* to:
   A. 2.44%.
   B. 3.44%.
   C. 3.66%.
5. Based on DDM, Sizemore's cost of equity is *closest* to:
   A. 11.39%.
   B. 13.82%.
   C. 14.42%.
6. How many of the factors considered by Hsei would support a higher specific company risk premium (SCRP) for Minicure?
   A. Only one of the factors.
   B. Two of the factors.
   C. All three factors.

KEY CONCEPTS

LOS 15.a

Top-down (i.e., macro) factors that affect the cost of capital include capital availability, market conditions, legal and regulatory considerations, and tax jurisdiction.

Bottom-up (i.e., company-specific) factors that affect the cost of capital include business or operating risk, asset nature and liquidity, financial strength and profitability, and security features.

LOS 15.b

If a company's debt is publicly traded, the yield to maturity for the firm's longest-maturity straight debt outstanding is our best estimate of its cost of debt. If a company's debt is not traded (or is thinly traded), we can use matrix pricing, based on the yields on traded securities with the same maturity and credit ratings. If the debt is not credit rated, we can use financial ratios of the company such as interest coverage or financial leverage to infer a credit rating on the debt.

For a finance lease, the rate implicit in the lease (RIIL) is the cost of debt. The RIIL can be estimated as the IRR that equates the fair value of the leased asset (plus the lessor's direct initial costs) to the present value of the lease payments plus the residual value.

For foreign borrower, a country risk premium (such as the yield difference between foreign sovereign debt and a benchmark government security) should be added.

LOS 15.c

There are two types of estimates of the equity risk premium: historical estimates and forward-looking estimates.

A historical estimate of the ERP consists of the difference between the historical mean return for a broad-based equity market index and a risk-free rate, over a given time period. Survivorship bias and non-stationarity in the time series are concerns with historical estimates.

Forward-looking estimates can be survey estimates, estimates based on DDM, or estimates based on macroeconomic variables.

Grinold-Kroner model: $\text{ERP} = [\text{DY} + \Delta\text{P/E} + i + \text{G} - \Delta\text{S}] - r_f$

LOS 15.d

Cost of equity based on DDM:

cost of equity (r_e) = dividend yield (DY) + capital gains yield (CGY)

Fama–French model: required return of stock = $r_f + \beta_1\text{ERP} + \beta_2\text{SMB} + \beta_3\text{HML}$

The five-factor Fama–French extended model adds two additional factors: profitability (RMW) and investment factor (CMA).

$$\begin{aligned}\text{required return of stock} = {} & r_f + \beta_1\,\text{ERP} + \beta_2\,\text{SMB} + \beta_3\,\text{HML} \\ & + \beta_4\,\text{RMW} + \beta_5\,\text{CMA}\end{aligned}$$

LOS 15.e

Required return for private companies can be calculated using an expanded CAPM that includes risk premiums appropriate to a private company: size premium (SP), industry risk premium (IP), and specific company risk premium (SCRP).

required return = $r_f + (\beta_{peer} \times \text{ERP}) + \text{SP} + \text{SCRP}$

Alternatively, the build-up approach adds to the risk-free rate the ERP, and any additional risk premiums as applicable, for size, industry, and company-specific characteristics.

required return = r_f + ERP + SP + IP + SCRP

LOS 15.f

This LOS uses a case study approach to pull together the concepts learned in the prior LOS.

For the exam, you should be prepared to evaluate a company's capital structure and its cost of capital relative to peers.

You may be asked to:

- Estimate a firm's after-tax cost of debt using a synthetic credit rating and an implied credit spread.
- Explain why a particular cost of equity is justified.
- Identify company characteristics that would justify a higher or lower SCRP.
- Estimate a firm's cost of equity using the extended CAPM and the build-up approach.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 15.1, 15.2

1. **A** Civil law countries afford fewer protections to investors as compared to common law countries and hence investors in civil law jurisdictions like Country Q may demand a higher risk premium. Countries with less-developed capital markets and highly volatile currencies would similarly call for a higher risk premium. (Module 15.1, LOS 15.a)

2. **A** Sizemore's interest coverage (IC) ratio = 3,421 / 814 = 4.20. Sizemore's financial leverage (D/E) ratio = 8,671 / 17,342 = 0.50. Both values fall within the range for BBB classification; this indicates a 1.20% spread. After-tax cost of debt = r_d (1 – T) = (2.30% + 1.20%) × (1 – 0.34) = 2.31% (Module 15.1, LOS 15.b, e)

3. **B** Statement 1 is incorrect. Arithmetic mean is a good estimate of a one-period expected return, but does a poor job of estimating multiperiod return (which determines expected terminal wealth). A geometric mean gives lower weight to outliers and estimates the expected terminal wealth more accurately.

   Statement 2 is correct. A weakness of the *historical approach* (and not forward-looking estimates) is the assumption that the mean and variance of the returns are constant over time (i.e., the ERP time series is stationary). *Historical estimates* (not forward-looking estimates) may suffer from survivorship bias. (Module 15.2, LOS 15.c)

4. **C** Expected inflation = (1+ nominal $\text{yield}_{\text{Treasury}}$) / (1 + $\text{yield}_{\text{TIPS}}$) – 1 = (1.0267 / 1.0033) – 1 = 2.33%. ERP = [DY + ΔP/E + *i* + G – ΔS] – E(r_f) = [1.10% – 0.10% + 2.33% + 3% – 0] – 2.67% = 3.66%

   Note: If a risk-free rate is separately provided, use that. In this case, we are given 10-year nominal Treasury yield, which in the absence of any other guidance is used as the risk-free rate. (Module 15.2, LOS 15.e)

5. **C** Using our calculator's CF function: CF0 = –25, C01 = 1, C02 = 1, C03 = 1 + 34, CPT IRR = 14.42% (Module 15.2, LOS 15.d)

6. **C** All three factors represent higher risk for investors in Minicure. Minicure's management team clearly has strong R&D skills, but that may not translate to experience and expertise in running a company. The assets of Minicure are not tangible; rather they are specialized and illiquid. The supplier concentration involved in the production of Mitra reduces the power of Minicure to extract a high margin for the product. (Module 15.2, LOS 15.d)

READING 16

CORPORATE RESTRUCTURING

EXAM FOCUS

For this reading, pay attention to the various types of restructurings, the motivations behind them, and their impacts on company value. The perspective here is that of an investment analyst seeking to identify the effect of an announced restructuring on company value. This reading is mostly qualitative: there is lots of terminology, and very little computation.

MODULE 16.1: RESTRUCTURING TYPES AND MOTIVATIONS

LOS 16.a: Explain types of corporate restructurings and issuers' motivations for pursuing them.

Large corporations often have diverse lines of business under a common corporate umbrella. This can yield benefits of group ownership called **synergies**. Synergies can be *cost synergies* (lower expenses), *revenue synergies* (increased sales), or a combination of the two. Lower costs can arise from economies of scale or elimination of redundant functions. Revenue synergies arise from economies of scope (e.g., cross-selling of products or reduced competitive pressure). Sometimes, inefficiencies or costs (i.e., negative synergies) may instead result from these diverse holdings due to diseconomies of scale, or loss of focus on core competencies.

Managers alter the portfolio of company businesses in response to a changing business environment to reduce inefficiencies and increase synergies. Figure 16.1 shows types of corporate transactions and company-specific motivations for each.

Figure 16.1: Types of Corporate Transactions

Transaction	Definition	Motivations
Investment	Increases the size of the company or the scope of the company's operations, thereby increasing company revenues and/or the revenue growth rate	Realize synergies, increase growth, improve company capabilities, acquire needed resources/talent, or acquire an undervalued target
Divestment	Reduces the size of the company by shedding parts of the business that have lower profitability, slower growth, or higher risk	Liquidity, valuation (i.e., fetching an attractive price), refocus on core business, or to comply with regulatory requirements
Restructuring	Does not change the size of the company, but improves its cost structure and capital structure to enhance profitability, increase growth, or reduce risk	Address financial challenges (including bankruptcy and liquidation), or improve return on capital

There are two *top-down* drivers of all three kinds of actions: high security prices overall and industry shocks.

High security prices. All three corporate transactions tend to be cyclical, increasing in prevalence during economic expansions and decreasing during contractions. There are several possible explanations for this:

- *Greater CEO confidence*. CEO confidence levels rise during rising markets when security prices are high; CEOs are more likely to execute major corporate actions in this environment.
- *Lower cost of capital*. When stock and bond prices are high, the cost of capital is low, allowing for less dilution to existing shareholders, or lower interest cost.
- *Overvalued stock*. It is attractive for the board and management to use an overpriced security in corporate transactions (e.g., to finance an acquisition).

Yet empirical studies suggest that corporate transactions taken during *weaker* economic times tend to create more value. A BCG study found that weak-economy deals tend to have a 10% higher rate of return than strong-economy deals over the three years following the transactions.

Industry shocks. Corporate restructuring also tends to have industry-specific waves: Mergers in an industry are often followed by more mergers. This phenomenon of **industry shocks** suggests a reactionary motivation behind some corporate transactions.

Within the three major categories of corporate transactions, there are nine specific types as shown in Figure 16.2.

Figure 16.2: Categories of Corporate Transactions

Investments	Equity investment
	Joint venture
	Acquisition
Divestments	Sale
	Spin-off
Restructurings	Cost restructuring ■ Outsourcing ■ Offshoring
	Balance sheet restructuring ■ Sale and leaseback ■ Dividend recapitalization
	Reorganization
Leveraged buyouts	

1. Investment Actions

Acquisitions can enable a company to expand quickly (e.g., into foreign countries) or to access inputs at a favorable price (vertical integration). There are three subtypes of investment actions:

a. A purchase of a material but less-than-50% stake in another company is an **equity investment**. While both companies maintain their independence, the investor may get board representation in the investee (depending on the size of the investment). An equity investment may represent a strategic investment (to influence the investee), a move toward an eventual acquisition, or simply an investment in a perceived undervalued asset

b. In a **joint venture**, two or more companies contribute resources, knowledge, or talent, and jointly control a separate, independent company. Often, a company seeking to expand into a foreign market may undertake a joint venture with a local company that has an existing distribution network and local know-how.

c. Controlling investments are **acquisitions** where the investee company becomes a subsidiary of the investor company. After the acquisition, the investor company reports consolidated financial statements, including the results of the investee company.

2. Divestment

Divestment is the action of selling off subsidiary business interests. Motivations for divestment include liquidity needs, regulatory requirements, or simply refocusing on core competencies. Companies may also sell divisions that have become overvalued by the market. Divestments can take two different forms: sales and spin-offs.

a. **Sale of a division** to another company is the opposite of an acquisition. After the sale, the seller company has no exposure to the divested business. The sale proceeds (i.e., capital freed up by divestiture) can be returned to shareholders or otherwise put to better use.

b. **Spin-off** involves creating a new, separate legal entity, and distributing equity in the newly created spin-off to the divesting company's shareholders. A spin-off is intended to improve the focus of the management and employees of the new company, and allows stock-based compensation schemes to be implemented. Unlike a sale, a spin-off does not generate any proceeds for the divesting company.

   The value of the divested business is often a key consideration in the choice between a sale and a spin-off: A moderate-to-large-sized business unit sought by several potential acquirers might be expected to fetch a high sale price.

3. Restructuring

A restructuring can be forced or opportunistic. In an opportunistic restructuring, a business changes its balance sheet composition, cuts costs, or alters its business model to improve the return on capital. A forced restructuring may be necessitated by overcapacity, poor management, falling demand, or a worsening competitive landscape.

a. **Cost restructuring** actions typically follow periods of underperformance, and pursue improvements in the operational efficiency of the company. Two common restructuring approaches are outsourcing and offshoring.

   i. **Outsourcing** involves contracting out standardized business process (e.g., payroll, human resources) to third-party vendors that have lower costs due to their economies of scale. A risk of outsourcing is the need to manage multiple contractual obligations.

   ii. **Offshoring** uses cheaper foreign labor while still keeping a business process in-house, and usually involves creating a wholly owned foreign subsidiary. Outsourcing and offshoring are often combined, with a business process outsourced to a foreign company.

b. **Balance sheet restructuring** involves changing the mix of assets, changing the capital structure, or both. Types of balance sheet restructuring include:

   i. **Sale and leaseback** involves selling an asset to a lessor, and then entering into a lease contract over the remaining economic life of the asset. The result is an immediate cash infusion for the seller. Lease payments should be higher than the depreciation of the asset, reflecting interest charged by the lessor. One motivation for a sale and leaseback transaction is if the lessor is able to obtain financing at more favorable terms than the lessee.

   ii. **Dividend recapitalization** involves increasing leverage on the balance sheet by increasing debt-financed dividends or by repurchasing shares. The objective is to replace equity in the capital structure with cheaper debt, thereby reducing the company's WACC. The higher risk that comes with increased leverage means that dividend recapitalization is most appropriate for issuers with stable cash flows.

c. **Reorganization** may be mandated by a court during an insolvency proceeding. Management's reorganization plan, which specifies the terms of exiting the bankruptcy proceeding, must be approved by the court. The court then oversees measures such as asset sales, refinancing, or conversion of debt to equity. If the court does not approve management's plan, the company may be liquidated.

4. Leveraged Buyout (LBO)

A **leveraged buyout (LBO)** is a special kind of corporate restructuring involving investment, divestment, and restructuring. In an LBO, a private equity firm first purchases a company using a large amount of debt to finance the transaction (the investment). Subsequently, unrelated parts are sold to generate cash to service the significant debt (the divestment). Reorganization of the remaining business operations is generally performed to generate synergies. Eventually there is an exit via a sale or public listing. If the target is an existing public company, an LBO may be referred to as a "take-private" transaction.

LOS 16.b: Explain the initial evaluation of a corporate restructuring.

The steps involved in analyzing an announced corporate action include:

1. Initial evaluation.
2. Preliminary valuation.
3. Modeling and valuation.
4. Update investment thesis.

Investment analysts seek to answer four questions when conducting the initial evaluation of an announced corporate restructuring: What, why, when, and is it material?

Corporate press releases, securities filings, and analyst conference calls help to answer the first three questions. While gathering information, analysts should attempt to understand the real motivations behind the restructuring, and contextualize it with their existing understanding of the company's business strategy.

Analysts typically evaluate materiality along the dimensions of size and fit. For an acquisition or divestiture, size is evaluated based on the value of the transaction (including cash, issuance of stock, or assumption of debt) as a proportion of the enterprise value of the acquiring company. Cost restructuring may be evaluated as estimated savings as a percentage of sales. In either case, a proportion greater than 10% generally indicates a material action.

Analysts should also evaluate a transaction in terms of fit with existing company strategy or with prior actions. For example, the divestment of an unrelated business for a company that had previously been diversifying into such businesses may indicate a change in strategy, or an acknowledgment that the strategy is not working.

The change in the company's stock price when a restructuring is announced is one indicator of the value that the market expects to be created or destroyed. This price change incorporates the market's estimate of the probability of the transaction actually completing (due to required shareholder/creditor/regulatory approvals, etc.) as well as the time the restructuring action will take. Note that the initial market reaction is often wrong when compared to the company's performance in subsequent years.

MODULE QUIZ 16.1

1. A large retail pharmacy chain has made a significant capital investment to incorporate health care kiosks within their stores. If the company would like to receive cash up front in

exchange for the kiosks, and yet retain the right for their future use, the company would *most* appropriately pursue a:

A. spin-off.
B. balance sheet restructuring.
C. reorganization.

2. A social media company has three business segments. While the “text messaging” and “video sharing” segments have a healthy growth rate, the “blogging” segment has been performing poorly due to changes in the regulatory environment. The *most* appropriate corporate action for this company would be a(n):

A. sale.
B. acquisition.
C. balance sheet restructuring.

3. Which of the following is *least* likely to represent a top-down driver of corporate actions?

A. Industry shocks.
B. CEO confidence.
C. High security prices.

MODULE 16.2: VALUATION

LOS 16.c: Demonstrate valuation methods for, and interpret valuations of, companies involved in corporate restructurings.

We now discuss *preliminary valuation* using relative valuation methods. For acquisitions or divestitures, we need to estimate the value of the target. There are three primary approaches to this valuation: discounted cash flow analysis, comparable company analysis, and comparable transaction analysis.

1. Discounted Cash Flow (DCF) Analysis

DCF analysis entails estimating the target’s future free cash flows using estimates of earnings, capital expenditures, working capital investments, and other inputs. These future free cash flows are discounted to generate an estimate of value. We will cover this in more detail in the Equity Valuation topic area.

2. Comparable Company Analysis

Comparable company analysis uses relative valuation metrics of comparable firms to estimate value, and then adds a takeover premium to determine a fair price for the acquirer to pay for the target.

Comparable company analysis (CCA) involves the following steps:

Step 1: *Identify the set of comparable firms*. CCA requires us to first select a set of other companies that are similar to the target firm. Ideally, the other companies will come from the same industry as the target and have a similar size, growth rate, operating margin, and ROIC.

Step 2: *Calculate various relative value measures based on the current market prices of companies in the sample*. Some analysts use relative value measures based on enterprise value (EV), which is the market value of the firm's debt and equity minus the value of cash and investments. These measures include EV to EBITDA, and less commonly EV to free cash flow. Equity multiples such as the price to earnings (P/E) ratio may also be used as relative value measures. Other sector-specific measures are also sometimes employed, such as EV to subscribers (for tech companies) or EV to reserves (for oil and gas companies).

Step 3: *Apply the average multiple to develop an estimated target value*. Applying the mean or median of the chosen multiples to the target provides an estimate of the value and allows for a basic reasonableness check for the transaction price announced.

Because the estimated value of the target under CCA does not include a control or takeover premium, CCA is most commonly used in the valuation of spin-offs rather than acquisitions.

Advantages of CCA:

- Data for comparable companies is easy to access.
- The assumption that similar assets should have similar values is fundamentally sound.
- Unlike a discounted cash flow approach, estimates of value are derived directly from the market rather than assumptions and estimates about the future.

Disadvantages of CCA:

- The approach implicitly assumes that the market's valuation of the comparable companies is fair.
- Comparable companies provide an estimate of a fair stock price, rather than a fair *takeover* price. An appropriate takeover premium must be determined separately.
- Sometimes the target may be unique and therefore there may not be any true peers.

3. Comparable Transaction Analysis

Comparable transaction analysis (CTA) is similar to CCA but uses *actual takeover transaction* prices rather than market prices of stock: in CTA, the comparables are actual takeover targets. Since the transaction price already includes a takeover premium, we don't have to estimate it separately to add to the CCA estimate of value.

Advantages of CTA:

- Since the approach uses data from actual transactions, there is no need to estimate a separate takeover premium.
- Unlike a discounted cash flow approach, estimates of value are derived directly from recent prices for actual deals completed in the marketplace, rather than from assumptions and estimates about the future.

Disadvantages of CTA:

- The CTA approach implicitly assumes that the M&A market valued past transactions appropriately. If past transactions were over- or underpriced, those mispricings will be carried over to the estimated value for the target.

- There may not be enough comparable transactions to develop a reliable estimate of the target value. An analyst who is unable to find enough similar companies might use M&A deals from other industries that are dissimilar to the deal being considered.
- Historical transactions may have occurred under different conditions (industry growth, regulatory frameworks, etc.) and hence may not represent current value reliably.

Premium Paid Analysis

In a takeover transaction, the acquiring firm typically pays a premium over the current market price as an incentive for the target to accept the offer. This premium can be estimated as:

premium = (DP – UP)/UP

where:

DP = deal price and UP = unaffected (i.e., pre-announcement) price

Some effect of pre-announcement rumors may be incorporated in the trading price of the target prior to the announcement; hence care should be taken when estimating UP. Analysts might use a week-old market price or volume-weighted trading price over the prior week as the UP to mitigate this issue.

LOS 16.d: Demonstrate how corporate restructurings affect an issuer's EPS, net debt to EBITDA ratio, and weighted average cost of capital.

We now turn to the modeling and valuation phase. The first step in this phase is to generate pro forma financial statements that reflect the impact of the corporate action. The steps in this process are:

Step 1: Combine acquirer and target revenues and make adjustments for revenue synergies or dissynergies.

Step 2: Combine operating expenses and make adjustments for cost synergies or dissynergies.

Step 3: Combine depreciation/amortization.

Step 4: Combine other income/expenses.

Step 5: Beginning with acquirer interest expense, add additional interest on debt issued to finance the acquisition.

Step 6: Estimate taxes based on the weighted average tax rates of the acquirer and the target.

Step 7: The number of shares outstanding should include the additional shares issued as consideration for the acquisition.

From these results, we can estimate the future EPS of the combined entity.

Estimating WACC

Restructuring can alter the cost of debt and equity capital. Several factors influence these costs: profitability (EBITDA to sales, or EBIT to sales), volatility of revenues or EBITDA, leverage (debt to EBITDA), collateral (asset specificity, liquidity, existence of an active market), and prevailing interest rates. Weights of debt and equity are calculated using market values, and include any financing raised or additional equity issued.

Using the data from pro forma financial statements, and the estimate of future WACC, the analyst can use discounted cash flow techniques to value the target.

MODULE QUIZ 16.2

1. Naomi Hirauye and Michael Klinkenfus, financial analysts with Mintier Textiles, are discussing potential methods of valuing a firm that Mintier is considering acquiring. As they are discussing the most appropriate valuation method to use, Klinkenfus makes two statements:

   Statement 1: One advantage of the discounted cash flow method is that it makes it easy to model changes in the target company's cash flow resulting from changes in operating synergies that may occur after the merger.

   Statement 2: Since the comparable transaction approach uses actual transaction data, there is no need to calculate a takeover premium.

   How should Hirauye respond to Klinkenfus's statements?

   A. Agree with both statements.
   B. Disagree with both statements.
   C. Agree with only one statement.

MODULE 16.3: EVALUATION

LOS 16.e: Evaluate corporate investment actions, including equity investments, joint ventures, and acquisitions.

We will illustrate the process of evaluating corporate investment actions using an equity investment as an example. (Similar processes can be used for other corporate actions.)

EXAMPLE: Evaluating an equity investment

Alpha, Inc., a large movie studio, is a mature business that is concerned about new competition eroding its market share. Beta LLC is a digital distribution company that provides proprietary entertainment content to its subscriber base. Beta has had high organic growth over the past three years, but needs an infusion of capital to fund continued growth. Beta has zero debt and one class of common shares. Selected financials of the two companies for the past year (trailing 12 months) are shown here.

Selected Financial Data for 20X1 (TTM) for Alpha and Beta (USD millions)

	Alpha	Beta
Net revenues	16,484	720
EBITDA	3,297	202
EBIT	2,112	57
Interest expense	–817	0
Income tax	–615	–12
Net income	680	45
Total debt	8,671	0

Alpha makes a $1 billion cash investment to acquire 25% of the equity in Beta. Alpha finances the equity investment by issuing senior unsecured debentures. The two companies enter into an agreement to distribute the content library of Alpha based on 50-50 revenue sharing.

An analyst gathers the following information related to the peers of Beta LLC:

Comparable Company Analysis for Beta (USD millions)

Comparable	Enterprise Value	Net Revenues (TTM)
P	1,431	311
Q	1,019	112
R	1,994	391
S	769	157

Evaluate the implied valuation of Beta and the transaction's effects on Alpha's solvency.

Answer:

Let's think about some of the questions that an analyst might want to consider:

1. *What other corporate actions could Alpha have undertaken?*

   Alpha alternatively could have secured a controlling interest in Beta (i.e., an acquisition). Alpha could also have entered into an agreement to set up a third separate company as a joint venture; however, because Beta needs an infusion of cash, Beta is more motivated to find an investor than to commit to a new investment.

2. *What are the primary motivations for the two companies to enter into the announced transaction?*

   Alpha will obtain entry into a fast-growing business segment, as well as gain a mechanism to monetize its assets (i.e., its content library). Beta will obtain the necessary infusion of cash to finance its growth.

3. *Evaluate the valuation of Beta implied by the transaction.*

   Given that Alpha acquired a 25% stake for $1 billion, Beta's implied market cap is estimated at $4 billion. Given that Beta has no debt, Beta's implied enterprise value is also $4 billion.

   The following table shows the EV/revenue multiples for the comparables as well as the mean and median multiples.

Comparable	Enterprise Value	Net Revenues (TTM) $M	EV/Rev
Beta LLC		720	
P	1,431	311	4.6
Q	1,019	112	9.1
R	1,994	391	5.1
S	769	157	4.9
Median			5.0
Mean			5.925

Using the mean EV/revenue multiple of 5.925, the EV of Beta would be estimated to be 720 × 5.925 = $4.27 billion. Using the median multiple, the EV of Beta would be estimated to be 720 × 5 = $3.6 billion. The median multiple may be more representative because the mean is significantly skewed by Comparable Q, which appears to be an outlier. At a justified value of $3.6 billion, the $4 billion purchase price appears to be at a premium of 11%.

Other things to take into account are risk and growth rate differences among the peer companies.

4. *What is the impact of the investment on Alpha's debt-to-net-income ratio, assuming no change in the financials for the two companies over the next year?*

   Alpha's current debt-to-net income = 8,671 / 680 = 12.75x.

   Alpha's 25% stake in Beta would increase Alpha's net income by 25% of the net income of Beta (this will be shown as "income from associates" in Alpha's income statement).

   Alpha's new net income = 680 + (0.25 × 45) = 691

   Alpha's new debt = 8,671 + 1,000 = 9,671

   Alpha's new debt-to-net income = 9,671/691 = 14x.

   Thus, one impact of the investment in Beta is that Alpha's debt-to-net income ratio will increase from 12.75x to 14x.

Joint Ventures

The process of evaluating a joint venture announcement is similar. Questions to consider include: What are the gains to the two firms involved? What impact will the transaction have on the various ratios of the two companies?

The accounting for a joint venture is similar to that of equity investments. The two partners in the joint venture will report their stake in the venture using the equity method, reporting their share of income from the venture in their respective income statements. Any capital raised by the two partners will be accounted for in their own financial statements (e.g., debt used to finance the investment in the joint venture would increase the debt level of the investor company).

Acquisitions

We cover acquisition accounting in detail in the Financial Statement Analysis topic area. The acquisition method calls for line-by-line consolidation of the investor's financial statements with the financial statement of the subsidiary, with a recognition of non-controlling interest in the consolidated financial statements.

Evaluating an acquisition should include determining the impact of estimated synergies on the profitability ratios. The impact of consolidation on leverage ratios (debt-to-equity or debt-to-EBITDA) should also be analyzed, to determine the potential for a change in the parent company's debt ratings (and hence, in its cost of capital). The evaluation should consider the impact of the transaction on the diluted EPS of the parent and whether the transaction price appears to be reasonable, using comparable transaction analysis and the premium-paid analysis discussed earlier.

LOS 16.f: Evaluate corporate divestment actions, including sales and spin offs.

Evaluation of divestment actions typically involves:

1. Valuation of business segments.
2. Impact on ratios.

For the sale of a business segment, analysts should form an opinion on whether the valuation of the segment is reasonable. One approach is to use a multiple such as EV-to-EBITDA or EV-to-sales to value the various segments of a business. The sum of the values of these individual segments is then compared to the "conglomerate" EV to determine if the market is properly valuing the sum of the parts.

When preparing pro forma financial statements, the proceeds from the sale (i.e., cash or stock of the purchaser and/or assumption of debt) should be accounted for properly. Ratios such as debt-to-EBITDA can then be prepared to evaluate the impact of the sale on the company's debt ratings.

Unlike sales, spin-offs do not generate sale proceeds, and hence are easier to model.

LOS 16.g: Evaluate cost and balance sheet restructurings.

Many external factors can be responsible for the initiation of a corporate restructuring, for example, the actions of an activist shareholder, a failed acquisition, etc.

The evaluation of a restructuring announcement involves preparing pro forma financial statements that reflect our expectations about cost savings, as well as assessing the likelihood of success in achieving the intended restructuring goals. Given a starting set of financial statements, you should be able to apply the given assumptions to generate pro forma values. These can then be used to calculate the relevant ratio(s).

MODULE QUIZ 16.3

Use the following information to answer Questions 1 through 4.

Fastnar is a large European industrial supplies company with a market capitalization of €17 billion. During the COVID-19 pandemic, Fastnar's supply chain experienced significant stress, leading to the loss of several large customers.

Fastnar's board has made the decision to acquire 100% of the outstanding stock of Luxor, a transportation company. Pre-announcement, Luxor's stock was trading at €260.

The pre-acquisition income statements (in € millions) for the two companies are shown here:

	Fastnar	Luxor
Revenues	28,695	8,964
Cost of goods sold	20,660	6,006
Operating expenses	2,583	807
EBITDA	5,452	2,151
Depreciation & amortization	1,121	1,671
EBIT	4,331	480
Interest expense	1,100	112
Tax	679	55
Net income	2,553	313
Shares outstanding (millions)	1,000	20
EPS	2.55	15.66

Additional details:

- A new issue of 30-year bonds yielding 3% will finance the purchase price of €6.26 billion.
- Synergies in the form of cost savings of €480 million per year are expected from this transaction.
- Additional depreciation due to the fair market value adjustment of Luxor's fixed assets will be €35 million per year.
- Excluding the items just mentioned, in the first year after the acquisition, the pro forma income statements are expected to be the same as the pre-acquisition income statements.
- Fastnar's tax rate is 21%.
- Analysts have identified the following comparables for Luxor and collected their P/E ratios based on TTM EPS as shown here:

Comparable	P/E
A	16
B	19
C	22
D	15

1. Relative to the pre-announcement price of Luxor, the acquisition premium paid by Fastnar is *closest* to:

   A. 12%.
   B. 14%.
   C. 20%.

2. Using the average P/E of the comparables, the acquisition price premium is *closest* to:

   A. 11%.
   B. 12%.
   C. 14%.

3. The consolidated EPS of Fastnar in the first year after acquisition is *closest* to:

A. €2.33.
B. €2.87.
C. €3.07.

4. Should an analyst consider this transaction to be material?

A. Yes, because of the size.
B. No, because of the size.
C. No, because of the fit.

Use the following information to answer Questions 5 through 7.

Super, Inc., is a large distributor of telecom equipment in South American markets. Several years ago, Super acquired Norse Telecom, a maker of low-cost switching equipment that was very popular in the emerging markets. Due to a recession in several key markets, Norse's revenue has declined significantly over the past several quarters. New competition has also squeezed Norse's profit margins. Unconsolidated financial data for the two companies is as follows.

Exhibit 1: Selected Financial Data (TTM) in USD millions

	Super	Norse
Revenues	18,132	8,113
EBITDA	2,176	568
EBIT	1,632	325
Debt	14,506	5,077

Super could potentially spin off the Norse investment by distributing Norse shares to Super's shareholders on a 1:1 basis. Alternatively, Super could sell Norse for $6 billion (the consideration includes the assumption of Norse's debt).

5. What would be the impact of spinning off Norse on Super's consolidated debt-to-EDITDA ratio? The ratio is *most likely* to:

A. stay the same.
B. increase.
C. decrease.

6. Assuming that the net proceeds from the sale would be used to pay down Super's debt, what would be the impact of selling Norse on Super's consolidated debt-to-EDITDA ratio? The ratio is *most likely* to:

A. stay the same.
B. increase.
C. decrease.

7. The median operating margin of Super's competitors is 11%. Super's management believes that a reorganization following the divestment of Norse can achieve this objective. The amount by which management will need to decrease Super's operating costs is *closest* to:

A. 363 million.
B. 1.286 billion.
C. 2.045 billion.

KEY CONCEPTS

LOS 16.a

Major corporate changes include investment (equity, joint venture, acquisition), divestment (sale, spin-off), and restructuring (cost and balance sheet).

Investment motivations include realizing synergies, increasing growth, improving company capabilities, acquiring resources and talent, or acquiring an undervalued target.

Divestment may be motivated by liquidity needs, high valuation, refocusing on a core business, or compliance with regulatory requirements.

Motivations behind restructuring may be financial challenges (including bankruptcy and liquidation) or simply to improve the return on capital.

LOS 16.b

The steps involved in analysis of an announced corporate action include:

1. Initial evaluation.
2. Preliminary evaluation.
3. Modeling and valuation.
4. Updating investment thesis.

Materiality is evaluated along the dimensions of size and fit. Transactions exceeding 10% of enterprise value, revenues, or market cap are considered material.

LOS 16.c

Comparable company analysis (CCA) uses relative valuation metrics for similar firms to estimate market value, then adds a takeover premium to determine a fair price for the acquirer to pay for the target. Since CCA does not incorporate the takeover premium directly, it is commonly used to value spin-offs as opposed to acquisitions.

Comparable transaction analysis (CTA) is similar to CCA but uses actual takeover transaction prices (as opposed to market trading prices). Since transaction prices already include a takeover premium, it is not necessary to estimate an add-on separately.

LOS 16.d

Evaluation of corporate transactions involve preparing pro forma financial statements that incorporate the terms of the transactions. Analysts should also estimate the impact of the transaction on the EPS, net debt-to-EBITDA ratio, and the WACC of the company.

LOS 16.e

In evaluating an equity investment, an analyst must recognize that income from associates is recorded in the investor's income statement. The consideration paid is used to adjust pro forma balance sheets.

A joint venture announcement is approached similarly: an analyst should calculate the gains to the firms involved, and the impact the transaction will have on the ratios of the two companies. Accounting for a joint venture is similar to that of equity investments.

Compared to other kinds of investments, acquisitions require a higher capital investment but allow for a controlling interest in the investee.

LOS 16.f

The focus here is on evaluating corporate divestment actions involving sales and spin offs.

This LOS is comprised of a case study of corporate restructuring that demonstrate the evaluation process, discussed in earlier LOS, undertaken by analysts upon their announcement. A strategic review is presented that evaluates the focus of a company, conglomerate discounts that may exist, and possible actions to realize value for its stakeholders.

LOS 16.g

This LOS demonstrates how to evaluate cost and balance sheet restructurings based on a case study that expands on and provides context for concepts introduced earlier. The example illustrates a cost restructuring that is prompted by a rejected acquisition offer and pressure from shareholders in response to that rejection. The case study demonstrates that there are multiple restructuring actions that can achieve the objective of increased shareholder value.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 16.1

1. **B** The company could unlock capital by undertaking a sale-and-leaseback transaction, which is considered to be a type of balance sheet restructuring. (LOS 16.a)

2. **A** Since the blogging business segment is underperforming, it may (absent synergies with other segments) be most appropriate to divest that segment. (LOS 16.a)

3. **B** CEO confidence is associated with the level of corporate actions, but is not itself a top-down factor. (LOS 16.a)

Module Quiz 16.2

1. **A** Hirauye should agree with both of Klinkenfus's statements. One of the key advantages to using the discounted cash flow method to value a target firm is that it makes it easy to model any changes that may result from operating synergies or changes in cash flow from the merger. One of the key advantages to the comparable transaction approach is that there is no need to compute a separate takeover premium as there is in the comparable company approach. (LOS 16.c)

Module Quiz 16.3

1. **C** Pre-announcement price per share = \$260 (given). Acquisition price per share = \$6,260 / 20 = \$313.

   Premium paid = (313 – 260) / 260 = 53/260 = 20.38%.

   (LOS 16.c, e)

2. **A** Average P/E of comparables: (16 + 19 + 22 + 15) / 4 = 18 Value of Luxor stock using average P/E: 15.66 × 18 = 282

   Premium = (313 – 282) / 282 = 33 / 280 = 10.99%

   (LOS 16.d, LOS 16.e)

3. **C** Consolidated net income before adjustments: 2,553 + 313 = 2,866

Adjustments:

Add synergies	480
Less: Additional depreciation	35
Less: Additional interest @3% of 6,260	188
Additional earnings (before tax)	257
Less: additional tax @ 21%	54
Additional net income	203

Adjusted consolidated net income: 2,866 + 203 = 3,069

Adjusted EPS = 3,069 / 1,000 = €3.07

(LOS 16.e)

4. **A** The acquisition value of €6.26 billion represents 37% of Fastnar's pre-acquisition market cap of €17 billion. A transaction greater than 10% of EV or market cap is considered material. The fit of the transaction may also indicate materiality as it suggests a shift in the business strategy to vertically integrate Fastnar's supply chain. (LOS 16.b, e)

5. **C** Current combined EBITDA = 2,176 + 568 = 2,744

Current combined debt = 14,506 + 5,077 = 19,583

Current debt-to-EBITDA ratio = 19,583 / 2,744 = 7.14

After the spin-off, Super's debt-to-EBITDA = 14,506 / 2,176 = 6.67

(LOS 16.f)

6. **C** Current combined EBITDA = 2,176 + 568 = 2,744

Current combined debt = 14,506 + 5,077 = 19,583

Current debt-to-EBITDA ratio = 7.14

Net proceeds from sale = 6,000 – 5,077 = 923

Repaying Super's debt to the extent of net proceeds, new debt = 14,506 – 923 = 13,583

New debt-to-EBITDA = 13,583 / 2,176 = 6.24

(Note: Since we are analyzing EBITDA, the interest savings do not factor in our calculation).

(LOS 16.f)

7. **A** 11% of current Super revenue = 0.11 × 18,132 = 1,995

Required increase of EBIT from current level = 1,995 – 1,632 = 363

(LOS 16.f)

TOPIC QUIZ: CORPORATE ISSUERS

You have now finished the Corporate Issuers topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow three minutes per question.

FORMULAS

Financial Statement Analysis

number of treasury shares = assumed proceeds ÷ average share price during the reporting period

assumed proceeds = cash proceeds + average unrecognized share-based compensation expense

U.S. GAAP interest cost = beginning PBO × discount rate

IFRS net interest income (expense) = beginning funded status × discount rate

funded status of the plan: funded status = fair value of plan assets − PBO

liquidity coverage ratio $= \dfrac{\text{highly liquid assets}}{\text{expected cash flows}}$

net stable funding ratio $= \dfrac{\text{available stable funding}}{\text{required stable funding}}$

underwriting loss ratio $= \dfrac{\text{Claims paid} + \Delta \text{loss reserves}}{\text{net premium earned}}$

expense ratio $= \dfrac{\text{Underwriting expenses including commissions}}{\text{net premium written}}$

loss and loss adjustment expense ratio $= \dfrac{\text{Loss expense} + \text{loss adjustment expense}}{\text{net premium earned}}$

dividends to policyholders ratio $= \dfrac{\text{Dividends to policyholders (shareholders)}}{\text{net premium earned}}$

combined ratio = loss and loss adjustment expense ratio + underwriting expense ratio

combined ratio after dividends = combined ratio + dividends to policyholders ratio

cash generated from operations (CGO)

= operating cash flow + cash interest + cash taxes

= EBIT + non-cash charges − increase in working capital

accruals^{CF} = NI − CFO − CFI

accruals ratio^{CF} $= \dfrac{(\text{NI} - \text{CFO} - \text{CFI})}{(\text{NOA}_{\text{END}} + \text{NOA}_{\text{BEG}})/2}$

PROFESSOR'S NOTE

Not all of the following ratios are used in this book. However, this list includes most of the common ratios that you are likely to encounter on exam day.

current ratio $= \dfrac{\text{current assets}}{\text{current liabilities}}$

quick ratio $= \dfrac{\text{cash}+\text{marketable securities}+\text{receivables}}{\text{current liabilities}}$

cash ratio $= \dfrac{\text{cash}+\text{short-term marketable securities}}{\text{current liabilities}}$

defensive interval ratio = (cash + short-term marketable investments + receivables) ÷ daily cash expenditures

receivables turnover $= \dfrac{\text{net annual sales}}{\text{average receivables}}$

inventory turnover $= \dfrac{\text{cost of goods sold}}{\text{average inventory}}$

days of sales outstanding (DSO) = average receivable collection period = $\dfrac{365}{\text{receivables turnover ratio}}$

days of inventory on hand (DOH) $(\text{DOH}) = \dfrac{365}{\text{inventory turnover}}$

payables turnover $= \dfrac{\text{purchases}}{\text{average payables}}$

number of days of payables $= \dfrac{365}{\text{payables turnover}}$

total asset turnover $= \dfrac{\text{net sales}}{\text{average total assets}}$

fixed asset turnover $= \dfrac{\text{net sales}}{\text{average fixed assets}}$

Corporate Issuers

effective tax rate = corporate tax rate + (1 − corporate tax rate)(individual tax rate)

expected increase in dividends = [(expected earnings × target payout ratio) – previous dividend] × adjustment factor

FCFE coverage ratio = FCFE / (dividends + share repurchases)

Grinold-Kroner model: ERP = [DY + ΔP/E + *i* + G – ΔS] – rf

Cost of equity based on DDM:

cost of equity (re) = dividend yield (DY) + capital gains yield (CGY)

Fama--French model: required return of stock = rf+ β1 ERP + β2 SMB + β3 HML

Five-factor Fama--French extended model:

required return of stock = rf + β1 ERP + β2 SMB + β3 HML + β4 RMW + β5 CMA

Expanded CAPM for private companies:

required return = rf + (βpeer × ERP) + SP + IP + SCRP

Build-up approach:

Required return = rf + ERP + SP + SCRP

Premium Paid Analysis premium = (DP – UP) / UP
= (deal price – unaffected price) / unaffected price

PROFESSOR'S NOTE
Not all of the following ratios are used in this book. However, this list includes most of the common ratios that you are likely to encounter on exam day.

gross profit margin $= \dfrac{\text{gross profit}}{\text{net sales}}$

operating profit margin $= \dfrac{\text{operating profit}}{\text{net sales}} = \dfrac{\text{EBIT}}{\text{net sales}}$

net profit margin $= \dfrac{\text{net income}}{\text{net sales}}$

return on assets $= \dfrac{\text{net income}}{\text{average total assets}}$

return on total capital $= \dfrac{\text{EBIT}}{(\text{interest bearing debt} + \text{shareholders' equity})}$

return on total equity $= \dfrac{\text{net income}}{\text{average total equity}}$

financial leverage ratio $= \dfrac{\text{total assets}}{\text{total equity}}$

long-term debt-to-equity ratio $= \dfrac{\text{total long-term debt}}{\text{total equity}}$

debt-to-equity ratio $= \dfrac{\text{total debt}}{\text{total equity}}$

debt-to-capital ratio $= \dfrac{\text{short-term debt} + \text{long-term debt}}{\text{short-term debt} + \text{long-term debt} + \text{total equity}}$

interest coverage $= \dfrac{\text{EBIT}}{\text{interest expense}}$

payout ratio $= \dfrac{\text{dividends paid}}{\text{net income}}$

retention ratio = 1 − payout ratio

earnings per share $= \dfrac{\text{net income} - \text{preferred dividends}}{\text{average common shares outstanding}}$

book value per share $= \dfrac{\text{common stockholders' equity}}{\text{total number of common shares outstanding}}$

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